A Latent Oscillator Measurement Model to Simulate Emotional-Expression Score Dynamics in Video

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Abstract

Machine-learning models for facial-expression recognition can turn video of a person into multivariate time series of classifier scores. These scores contain measurement error from model limitations, video quality, and observation conditions. They therefore cannot establish whether channels labeled as positive or negative emotion reflect lower-dimensional latent expressive processes, or whether an analysis method can recover those processes. We introduce the Latent Oscillator Measurement Model (LOMM), a measurement-aware simulation environment that separately specifies oscillator-based latent dynamics, time-varying activity, and an observation model with fallible indicators. In Study 1, a five-fold cross-fitted exploratory comparison evaluated simulated and real score series from 100 facial-expression video clips. Allowing clips to differ in baseline and scale reduced the selected composite discrepancy by 21.0%, from 0.3250 to 0.2567. Study 2 evaluated four methods used in network psychometrics and intensive longitudinal analysis through a common network-community recovery target using synthetic continuous indicators. At 100 observations per clip, with failed or timed-out attempts counted as incorrect, overall correct-dimension recovery was 0.939 for DynEGA, 0.884 for static EGA, 0.777 for GraphicalVAR, and 0.176 for GIMME. Independent starting states reduced recovery for all three methods tested, indicating that common-initial-state results were optimistic. LOMM provides a way to test whether methods recover intended structure before empirical score patterns are given psychological meaning.

Introduction

Facial-expression classifiers can turn video of a person into multivariate time series of classifier scores. Each frame yields a vector of scores for a chosen label set. These series have a temporal resolution that manual coding cannot sustain.

Earlier systems usually returned scores for categories inherited from their training data, whereas CLIP-based vision-language models can score user-specified labels (Foteinopoulou & Patras, 2023; Li et al., 2023; Radford et al., 2021). The scores are measurements produced by a model, a prompt set, and a video-processing pipeline. Their values also depend on image quality and observation conditions. Benchmark performance varies with model adaptation, label choice, and evaluation data (Foteinopoulou & Patras, 2023; Li et al., 2023). An interpretable pattern among score channels may therefore reflect expressive behavior, the measurement process, or both. The eleven-channel procedure examined here is treated as a source of classifier scores, without a frame-level claim that its labels measure subjective emotional experience.

This creates a psychometric problem. Several score channels may be analyzed separately, or they may be treated as fallible indicators of fewer latent expressive processes (Mehu & Scherer, 2015; Nesselroade et al., 2002; Russell, 2003). Measurement error can change the structure recovered by dynamic methods (Kankaanpää et al., 2026). Empirical data alone cannot show whether a method would recover the number of dimensions and the assignment of indicators to dimensions under a known measurement structure.

We introduce the Latent Oscillator Measurement Model (LOMM) to make that comparison possible. LOMM is a measurement-aware data generator for multivariate intensive longitudinal data. It simulates latent processes that change over time, specifies how strongly each process is expressed at each observation, and converts those processes into fallible observed indicators. The number of dimensions, indicator assignments, dynamics, activity pattern, loadings, and measurement error are specified before a recovery method is applied. LOMM combines oscillator models of intraindividual dynamics with a factor-analytic observation model (Boker, 2015; Boker & Nesselroade, 2002; Nesselroade et al., 2002; Zhang et al., 2008).

We use LOMM in two studies. Study 1 is a five-fold cross-fitted exploratory comparison of simulated scores with classifier scores from 100 MAFW videos. It asks whether eight descriptive features of the simulated and real score series are similar. Study 2 evaluates four methods used in network psychometrics and intensive longitudinal analysis: Dynamic Exploratory Graph Analysis (DynEGA), static EGA, GraphicalVAR, and GIMME. It asks whether they recover a factor structure used to generate synthetic continuous indicators. Static EGA provides a deliberately time-agnostic and computationally light benchmark, showing how much structure is recoverable from contemporaneous indicator levels alone. Study 2 also tests two consequential boundaries: whether a common initial state makes recovery easier and whether the original indicator grouping survives transformation to bounded scores. Because the empirical segments are short, the Study 2 results include a $T = 25$ panel as an ecological-length check rather than a second primary design. Figure 1 summarizes the distinct questions.

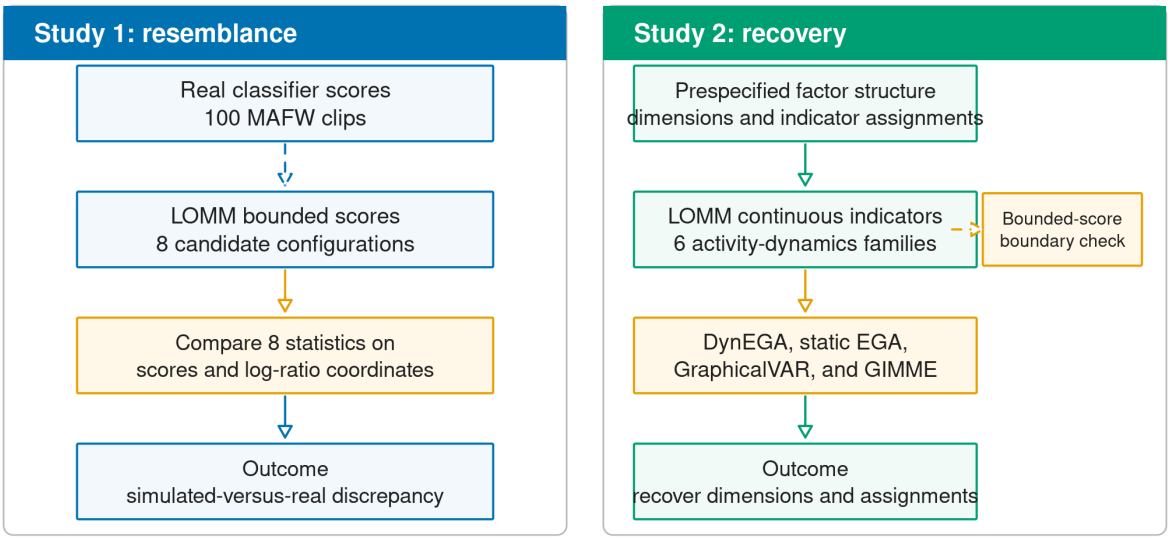


Figure 1: The two uses of LOMM in this paper. Study 1 compares descriptive features of simulated bounded scores with classifier scores from MAFW videos. Study 2 compares the dimensionality and indicator assignments recovered by four methods with the factor structure used to generate synthetic continuous indicators. A targeted bounded-score condition tests whether that original grouping survives transformation to scores that sum to one.

The Latent Oscillator Measurement Model

LOMM represents a familiar psychometric situation: several observed channels are fallible indicators of fewer latent processes, and those processes change over time. Consider a simulation with three expressive processes: approach-affiliation, threat-defense, and orienting-novelty. Anger, fear, disgust, and anxiety could be assigned as indicators of threat-defense. Their observed scores would share movement from that process while retaining channel-specific measurement error. A successful recovery method should place those indicators together and recover all three dimensions. These labels illustrate a possible simulation design; validation of corresponding affect categories in MAFW lies outside this study.

LOMM generates data in three layers. The latent layer describes how each process changes. The activity layer determines how strongly that process is expressed at each observation. The observation layer creates multiple indicators with prespecified loadings and measurement error. Keeping these layers separate allows a researcher to change the dynamics, expression pattern, or measurement quality while holding the other components fixed.

For recording i and dimension f, let $x_{if}(t)$ denote the latent level and $\dot{x}_{if}(t)$ its rate of change. LOMM treats each dimension as a linear second-order oscillator:

$$\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}, \qquad t \in [t_n, t_{n+1}). \tag{1}$$

The position-feedback parameter η_f governs the pull toward rest, and the velocity-feedback parameter ζ_f governs damping or amplification. The acceleration input $q_{if,n}$ adds irregular peaks and reversals during the interval between observations n and $n+1$. Negative η_f and ζ_f produce asymptotic return, with oscillation when the characteristic roots are complex. Setting $\zeta_f = 0$ with $\eta_f < 0$ produces undamped oscillation, while positive ζ_f in the tested conditions produces finite-horizon amplification. Appendix A gives the exact transition used to advance this system between observations, the stability conditions, and the initialization rules.

The activity layer multiplies the latent trajectory by a nonnegative value, $\tilde{x}_{if,n} = a_{if,n} x_{if,n}$. A process may remain strongly expressed, remain near a background level, or alternate with other processes. This layer matters because an abrupt change in activity can create a large observed change even when the underlying oscillator has not accelerated.

The observation layer is an ordinary factor model:

$$\mathbf{z}_{in} = \Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in}, \qquad \varepsilon_{in} \sim \mathcal{N}(\mathbf{0}, \Sigma_i). \tag{2}$$

Each row of Λ_i contains an indicator's loadings on the latent dimensions, and ε_{in} represents measurement error in the observed channels. Some configurations also allow recordings to differ in their overall baseline and scale. Study 1 estimates those components from training videos. The primary Study 2 design removes them while retaining small recording-specific differences in loadings and residual variances. Appendix A gives the complete observation equation and each construction.

LOMM supports two observed data forms. The primary Study 2 designs retain the continuous indicators $\mathbf{z}_{in}$, so the factor structure used to generate the data is defined directly on the variables being analyzed. Study 1 maps lower-dimensional coordinates to positive scores that sum to one, matching the form of classifier output. A targeted Study 2 condition applies the same type of bounded transformation to the continuous indicators and tests whether their original grouping survives. Appendix A gives the softmax, log-ratio, and identification details.

In this paper, LOMM supplies datasets with known dimensionality and indicator assignments. It is not fitted to the MAFW videos. Estimating oscillator, activity, and measurement parameters from empirical data

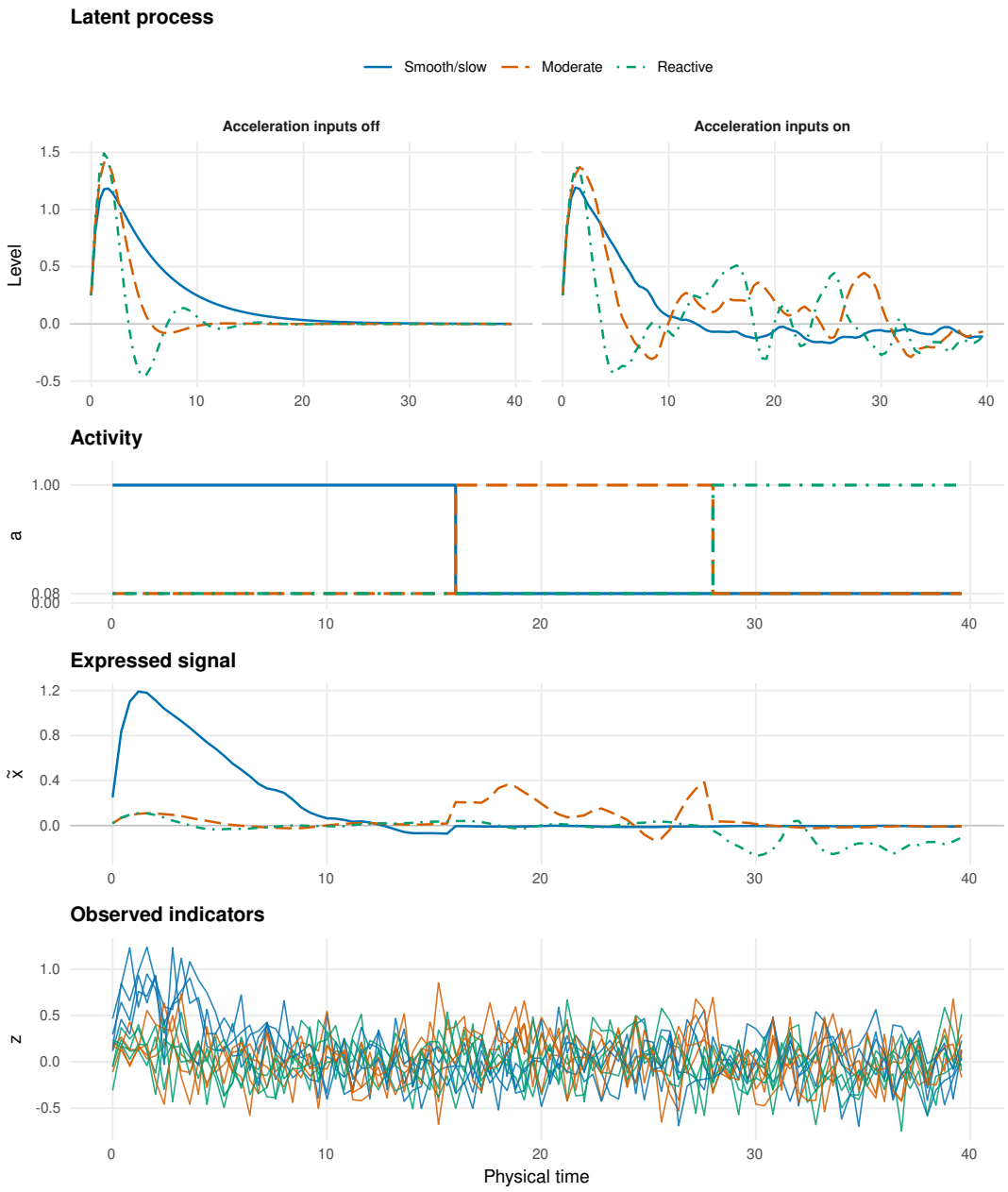


Figure 2: A worked LOMM example. The latent panel shows the same oscillator with acceleration inputs off and on. Activity determines how strongly the process is expressed, their product is the expressed signal, and the observation model produces grouped indicators that share that signal plus measurement error.

would require additional sign, scale, temperature, and frequency constraints, which are stated in Appendix A.

Study 1: Comparing Simulated and Real Score Series

Study 1 asks whether LOMM can reproduce selected features of real classifier-score series. We analyzed 100 naturalistic clips from the *MAFW: A Large-scale, Multi-modal, Compound Affective Database for Dynamic Facial Expression Recognition in the Wild* dataset (Liu et al., 2022). The transforEmotion package scored up to 50 frames per clip with the `oai-large` setting, which uses the OpenAI CLIP ViT-Large/14 model family (Tomašević et al., 2026). Eleven MAFW labels were passed directly to the text encoder. These channels are model outputs. The dataset and scoring procedure do not validate the labels as measures of subjective emotion.

Figure 3 shows the empirical data used as the comparison target. The main analysis contains 3,557 observations in 188 uninterrupted clip segments from the 100 videos. Segment lengths range from 16 to 50 observations, with a median of 35. Because these records are short, autocorrelation and spectral summaries require caution. A sensitivity analysis interpolated internal gaps of at most two frames. Appendix B gives the face-processing, score-validity, missing-data, and sampling details.

Study 1 compares eight LOMM configurations formed by including or excluding three features: clip-specific baseline, clip-specific scale, and a brief transition in activity. The comparison uses eight summary statistics. Six describe each channel over time: median level, variability of levels, variability of observation-to-observation changes, lag-1 autocorrelation, entropy of monotonic run lengths, and normalized spectral entropy. Effective rank and total variance describe the channels jointly. We evaluate all eight statistics on both the bounded scores and their log-ratio coordinates.

For each statistic, discrepancy combines the standardized difference in medians with the Wasserstein distance between the simulated and real distributions across clips. The larger distance is retained, then the eight statistics are averaged within each representation and the score and log-ratio representations receive equal weight. Smaller values indicate closer agreement.

The videos were divided into five folds. Measurement parameters and candidate settings were estimated or selected using the training clips, and each selected configuration generated 100 datasets matching the held-out clips' observation indices and segment structure. The held-out folds were used to compare the eight configurations and select the best one, so this is a cross-fitted exploratory comparison rather than external validation. Appendix B gives the complete discrepancy, calibration, search, and bootstrap procedures.



Figure 3: Score series from one MAFW video clip. Panel A shows four frames reproduced to clarify the academic analysis under the MAFW research-use terms. Panel B shows the classifier scores assigned to those frames. Panel C shows smoothed trajectories for the channels with the highest mean scores. Scores were generated with transforEmotion using OpenAI CLIP ViT-Large/14.

Study 2: Recovery of Dimensionality and Indicator Assignments

Study 2 asks whether four network-based procedures recover a prespecified number of dimensions and indicator assignments. The primary analyses use synthetic continuous indicators. Each dataset contains $p = mk$ indicators, where m is the number of dimensions and k is the number of indicators per dimension. Clips share the groups of indicators measuring the same latent dimension and the nominal settings, with small clip-specific loading and uniqueness jitter. Table 1 lists the six activity-dynamics families. Appendix B gives their exact schedules and parameters.

Every method receives the same simulated dataset. Each method estimates its own structure, that structure is converted to an undirected network by a prespecified rule, and the same community-detection procedure identifies groups of indicators. The estimated grouping is then compared with the assignment used to generate the data. This creates a common recovery target while preserving the statistical model and row handling of each procedure.

Table 1: The six Study 2 scenario families. Numeric activity gains, dwell fractions, and oscillator parameters are in Appendix B.

Code	Family
A	Weak expression, one dominant dimension
B	Strong expression, one dominant dimension
C	Strong expression, switching dimension
D	Rapid switching
E	Growing expression, one dominant dimension
F	Growing expression, switching dimension

Table 2: The four Study 2 procedures and the structure used for the common grouping target.

Method	Structure used for grouping	Role in the comparison
DynEGA	Partial-correlation network of estimated changes	Focal dynamic grouping method
Static EGA	Partial-correlation network of contemporaneous levels	Time-agnostic, computationally light benchmark
GraphicalVAR	Contemporaneous residual partial-correlation network	Lagged network-model comparison
GIMME	Average residual-covariance network across clip-level fits	Feasibility-stress comparison

DynEGA is the focal method because it groups indicators whose estimated changes are coordinated over time (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017). GraphicalVAR separates lagged relations from contemporaneous relations and supplies the latter for grouping (Epskamp et al., 2018; Epskamp, 2024). GIMME fits each clip separately; we average residual covariance matrices from successful fits before grouping (Gates & Molenaar, 2012; Lane et al., 2019). This use places GIMME under unusual computational strain, so its findings concern the executed comparison and should not be generalized to idiographic applications.

Static EGA applies EBICglasso and the same grouping step to the contemporaneous correlation matrix, without using the observation order. It provides a direct statistical baseline: if the prespecified groups are already visible in contemporaneous levels, a dynamic model may not improve this recovery target. It also provides a technical baseline because it requires much less computation. Comparing static EGA with DynEGA and GraphicalVAR therefore shows whether modeling temporal structure improves grouping enough to justify its cost in these conditions.

The primary $T = 100$ design contains 21,600 datasets. It crosses six scenario families, two or three dimensions, 50 or 100 clips, three loading levels, three residual-error levels, and 100 paired replications, with four indicators per dimension. Every method receives the same dataset within a condition and replication. Because MAFW segments are short, with a median of 35 observations, a $T = 25$ version of the same conditions is reported with the recovery results rather than as a co-equal design. Louvain at resolution 1.0 is the primary grouping algorithm; Leiden and Walktrap are paired sensitivity analyses applied to the same fitted networks.

The headline outcome is the overall correct-dimension recovery rate, P(valid output and correct dimension). It counts a failed or timed-out attempt as an unsuccessful recovery. We also report P(valid output) and P(correct dimension | valid output) so computational feasibility can be separated from accuracy among valid outputs. Assignment accuracy, adjusted Rand index, and variation of information assess the complete indicator partition among valid outputs. Trajectory-summary diagnostics for recovered-group summaries are reported in Appendix C; they are not Study 2 estimands.

Three additional analyses test scope. First, paired targeted departures alter starting states, activity transitions, cross-loadings, residual dependence, input dependence, or the observed scale one feature at a time. The bounded-score departure applies softmax directly to the p continuous indicators and analyzes both the raw scores and their centered log-ratio (CLR) components. This transformation changes the target geometry, so the original indicator grouping need not be preserved. Second, a restricted analysis adds clip-specific baseline and scale differences within the loading space. Third, an extended 518,400-dataset factorial simulation evaluates DynEGA over additional dimensions, indicators per dimension, and observation schedules. Appendix B gives the exact method settings, departures, outcome construction, resampling, and computing environment.

Results

Study 1 Results: Similarity of Simulated and Real Score Series

Figure 4 shows the score-scale discrepancy for all eight configurations. The *common baseline-and-scale configuration* gives every clip the same baseline and scale. The *baseline-only* and *baseline-and-scale* configurations allow clips to differ in level or in both level and variability. The remaining configurations add a brief activity transition.

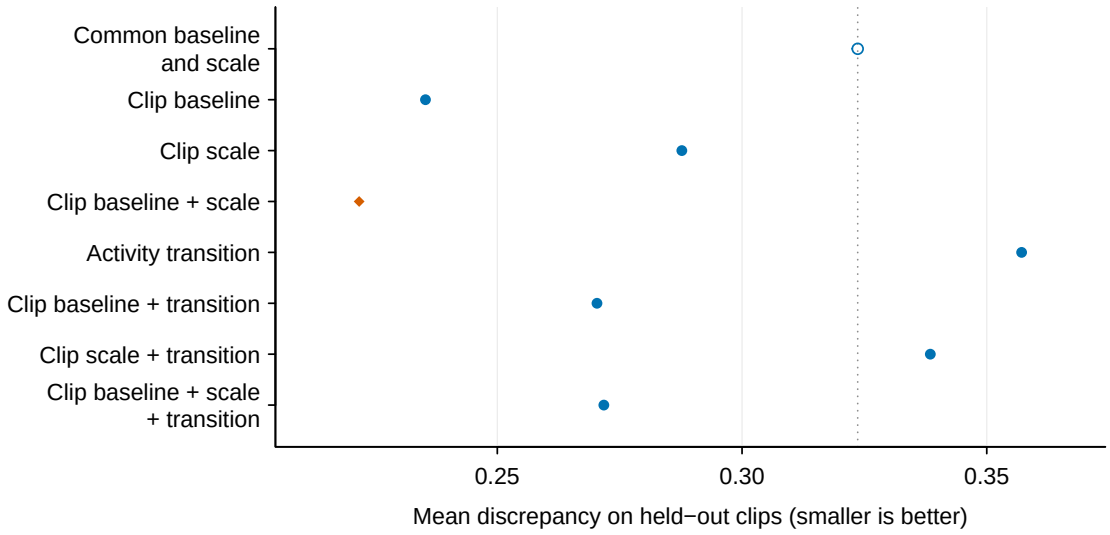


Figure 4: Score-scale discrepancy between simulated and real series for the eight configurations in the five-fold cross-fitted exploratory comparison. Points average the eight summary-statistic discrepancies; smaller values indicate closer agreement. The diamond marks the baseline-and-scale configuration. The composite used for selection weighted this score representation and the log-ratio representation equally. Appendix C reports both representations and the bootstrap diagnostics.

Averaging the score and log-ratio representations, the baseline-and-scale configuration had a discrepancy of 0.2567, compared with 0.3250 for the common baseline-and-scale configuration. Allowing clips to differ in baseline and scale therefore reduced the selected composite discrepancy by 21.0%. The baseline produced most of the improvement, the scale helped less, and adding a brief activity transition worsened the fit. Interpolating short internal gaps preserved the ordering: the baseline-and-scale discrepancy was 0.2511, compared with 0.3187 for the common configuration and 0.2707 for baseline only.

Important mismatch remained. The largest discrepancies concerned total score variance and the variability of observation-to-observation changes in log-ratio coordinates. The same held-out folds were used to compare configurations and select the best one, so these findings are exploratory. They support baseline and scale variation as a useful starting point for simulating these score series, without validating the classifier labels or LOMM as a model of emotional expression. Appendix C reports every configuration, statistic, bootstrap diagnostic, and the closest of 100 generated realizations for the example clip.

Study 2 Results: Recovery of Dimensionality and Indicator Assignments

Table 3 reports the $T = 100$ Louvain results. The overall recovery rate requires a valid network output and the correct number of dimensions, so every failure and timeout counts as an unsuccessful recovery. The final

two columns add assignment accuracy and computation time to the dimensionality results.

Table 3: Results for the primary $T = 100$ simulation design under Louvain. Assignment accuracy conditions on a valid network output. Runtime is the median among valid network outputs and includes the fit-to-grouping workflow.

Method	Valid output	Correct given valid	Overall recovery	Assignment accuracy	Median runtime, s
DynEGA	1.000	0.939	0.939	0.965	3.69
Static EGA	0.997	0.887	0.884	0.922	2.15
GraphicalVAR	0.941	0.826	0.777	0.902	85.41
GIMME	0.224	0.783	0.176	0.877	518.16

DynEGA returned a valid network for all 21,600 datasets and had the highest overall correct-dimension recovery rate, 0.939. Static EGA followed at 0.884, showing that much of the grouping was visible in contemporaneous indicator levels. GraphicalVAR reached 0.777. GIMME recovered the correct dimension in 0.783 of its valid outputs, but returned a valid network for only 0.224 of datasets, reducing its overall recovery to 0.176. Conditional assignment accuracy was 0.965 for DynEGA, 0.922 for static EGA, 0.902 for GraphicalVAR, and 0.877 for GIMME. Static EGA was about 40 times faster than GraphicalVAR and had higher overall recovery and assignment accuracy for this grouping target. GraphicalVAR estimates a richer time-series model, so this result concerns the value of that additional structure for the stated target and conditions. Appendix C reports Monte Carlo errors, intervals, and the remaining partition diagnostics.

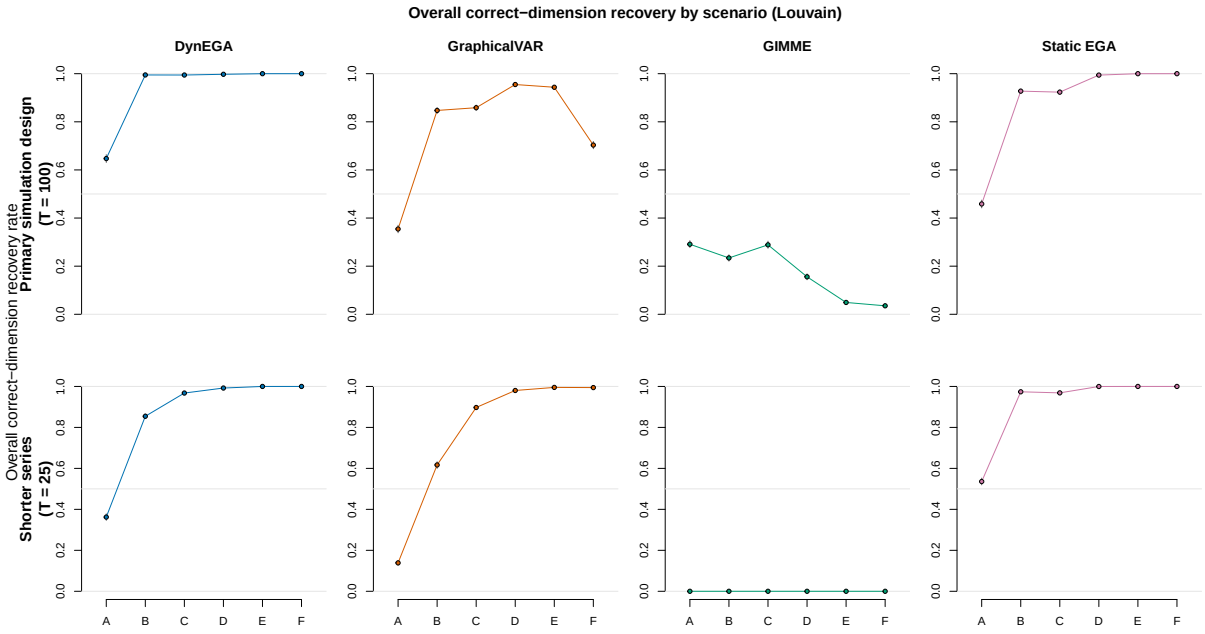


Figure 5: Overall correct-dimension recovery under Louvain across the six scenario families, with 25 or 100 observations per clip. A recovery requires a valid network output and the correct number of dimensions. Intervals resample whole datasets. Scenario codes are defined in Appendix B.

Recovery varied across scenarios and record lengths (Figure 5). DynEGA was strongest in most conditions. Static EGA's high recovery reflects block structure visible in contemporaneous levels. GraphicalVAR struggled most when expression or measurement quality was weak, and GIMME's low overall rates largely reflected missing valid outputs. In the extended factorial simulation, DynEGA recovered the correct dimension in 93.7% of 518,400 datasets and assigned 95.2% of indicators correctly. Stronger loadings and smaller residual error improved recovery. Leiden and Walktrap reproduced the Louvain interpretation.

Among correct-dimensional valid outputs at $T = 100$, trajectory correlation and NRMSE were 0.574 and 1.418 for DynEGA, 0.791 and 0.568 for GraphicalVAR, 0.606 and 0.898 for GIMME, and 0.787 and 0.563 for static EGA. These are post-estimation summaries of recovered groups. They show that grouping recovery and trajectory alignment are distinct targets and do not provide a general method ranking. Appendix C reports the complete diagnostics.

Robustness and Sensitivity Analyses

With $T = 25$, static EGA had the highest overall correct-dimension recovery rate at 0.913, followed by DynEGA at 0.863 and GraphicalVAR at 0.770. Shorter records modestly reduced DynEGA recovery, left

GraphicalVAR similar, and improved static EGA. Under the specified procedure and 900-second limit, no $T = 25$ GIMME attempt yielded a valid network output.

Independent starting states produced the largest targeted change. Overall correct-dimension recovery fell from 0.993 to 0.838 for DynEGA, from 0.912 to 0.525 for static EGA, and from 0.683 to 0.253 for GraphicalVAR. The common-initial-state design was therefore optimistic for all three methods.

Gradual activity transitions, small cross-loadings, and correlated inputs changed little in the tested conditions. Residual equicorrelation of 0.10 across all indicators increased recovery for GraphicalVAR and static EGA, but also created associations between dimensions and therefore has no general favorable interpretation.

Bounding the scores changed the question being asked and sharply reduced overall correct-dimension recovery on the original channels. DynEGA recovered the correct number of dimensions in 0.333 of the CLR datasets and 0.233 of the raw bounded-score datasets; the corresponding figures were 0.333 and 0.167 for static EGA, and 0.087 and 0.002 for GraphicalVAR. Direct softmax normalization and its accompanying projection to the zero-sum subspace did not preserve the structure among the original channels, and these results are the reason Study 1 and Study 2 have to stay separate.

Adding clip-specific baseline and scale differences in selected stable conditions changed DynEGA from 0.775 to 0.750, a paired difference of -0.025 with a 95% interval of $[-0.100, 0.050]$. Static EGA rose from 0.275 to 1.000 because the added baseline lay in the loading space and strengthened the intended grouping. GraphicalVAR and GIMME were unchanged in this small focused analysis, but their estimates had little support: GraphicalVAR produced only three correct outputs among 80 datasets, and every three-dimensional GIMME attempt timed out.

Median runtime among valid $T = 100$ outputs was 2.15 seconds for static EGA, 3.69 for DynEGA, 85.41 for GraphicalVAR, and 518.16 for GIMME. GraphicalVAR timed out on 1,271 of 21,600 attempts. GIMME produced 4,846 valid outputs, 14,477 timeouts, and 2,277 post-estimation errors. DynEGA returned a valid output for every primary and extended-factorial attempt. These values describe the executed workflows on one machine. Appendix C gives the full runtime, factorial, and community-detection results.

Discussion

LOMM addresses a common measurement problem in behavioral research. A classifier can produce multivariate time series from video, and an analysis can return dimensions or networks from those series, yet the empirical data alone cannot reveal whether the intended structure was recoverable. LOMM makes the

recovery target explicit by specifying latent dynamics, time-varying activity, and the observation model separately. This separation is the paper's central contribution. It also applies beyond video whenever intensive longitudinal measures are treated as fallible indicators of lower-dimensional psychological processes. CLIP is external to LOMM and supplies one empirical scoring pipeline for Study 1.

Study 1 evaluated resemblance to that empirical pipeline. Allowing clips to differ in baseline and scale reduced the selected discrepancy by 21.0%, with most of the improvement coming from the baseline. This result is local to 100 MAFW clips, one prompt set, one scoring model, and the chosen summary statistics. The same held-out folds supported comparison and selection, making the analysis exploratory. The result supports a simulation starting point for these score series and provides no validation of the emotion labels or of LOMM as a substantive theory of expression.

Study 2 evaluated recovery from synthetic continuous indicators with a prespecified factor structure. DynEGA achieved the highest overall correct-dimension recovery and was the only method evaluated in the extended factorial simulation. Static EGA showed that much of the grouping was recoverable from contemporaneous levels. It exceeded GraphicalVAR in dimensional recovery and assignment accuracy for this target while requiring a median 2.15 seconds rather than 85.41 seconds. This accuracy-cost result is substantively useful: temporal modeling did not improve the stated grouping target enough to offset its added cost in these conditions. GraphicalVAR addresses additional temporal questions, and the comparison does not rank those broader capabilities.

The trajectory-summary diagnostics in Appendix C reinforce the importance of defining the target. DynEGA's advantage concerned recovery of the prespecified dimensionality and indicator partition. Alignment of recovered-group summaries with latent trajectories produced a different ordering. Dimensional recovery and trajectory alignment should therefore be evaluated separately. All comparisons apply to the executed estimation, network-conversion, and grouping procedures. GIMME's low overall recovery mainly reflected its difficulty returning valid outputs under an unusual clip-level fitting procedure.

Initialization placed the strongest qualification on the primary design. A common starting state creates a shared transient across dimensions, which adds recoverable structure outside the measurement partition. Independent starting states reduced recovery for DynEGA, static EGA, and GraphicalVAR, showing that the common-initial-state design was optimistic. Gradual activity transitions, small cross-loadings, and correlated acceleration inputs changed little in the tested conditions. The bounded-score departure produced a sharper boundary: softmax normalization changed associations among the original indicators, and neither raw bounded scores nor CLR components preserved their original grouping. The main Study 2 recovery

claim therefore applies to synthetic continuous indicators.

Several limitations define the next tests. The latent dimensions are uncoupled. The finite-horizon amplification families produce large late-record amplitudes and serve as stress tests. The $T = 300$ conditions also change the observation interval, so they do not isolate record length. The GIMME residual-covariance conversion was constructed for this comparison and differs from typical idiographic use. Study 1 uses short clips from one controlled-access dataset and one zero-shot scoring pipeline. LOMM is used here as a generator with prespecified parameters; empirical estimation would require additional identification constraints.

Three practical lessons follow. Initial conditions can create structure unrelated to the intended measurement partition. A bounded transformation can change the target among the analyzed variables. Computational feasibility belongs in recovery evaluation because a method that does not return an output cannot recover the target. Applied work should therefore specify the measurement, latent dynamics, activity process, observation model, and recovery target before interpreting an empirical network. LOMM provides an environment for that sequence and can be extended to other scoring systems, coupled latent dimensions, and targets defined directly for bounded data.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study used simulations and secondary video data obtained under the MAFW research-use terms. It involved no new recruitment, intervention, or direct collection of human-participant data by the authors.

Consent to participate

Not applicable.

Consent for publication

The manuscript includes four still frames from MAFW solely to illustrate the score-extraction procedure. The frames are cited to the dataset, and no source video is redistributed. Their inclusion follows the dataset's terms for limited reproduction in academic publications.

Availability of data and materials

The project GitHub repository at https://github.com/atomashevic/LOMM contains the simulation summary results, analysis-ready tables underlying the figures, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison. Raw MAFW videos and frame-level classifier scores are not redistributed. Access to MAFW is governed by the dataset provider, and an end-to-end reanalysis of Study 1 requires approved access and local generation of the classifier scores.

Code availability

Code for generating LOMM datasets, fitting the methods, reproducing the statistical summaries, and regenerating the tables and figures is available in the same GitHub repository. The repository also provides the manuscript source and instructions for reproducing the analyses.

Authors' contributions

Aleksandar Tomašević contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

Open Practices Statement

The study was not preregistered. The simulation and analysis code, manuscript source, paper-facing tables, and aggregate MAFW-derived summaries used in the simulated-versus-real comparison are available at https://github.com/atomashevic/LOMM. Raw MAFW videos are third-party materials, and the frame-level classifier scores are derived from those restricted materials; neither is included.

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Appendix A: Complete Specification of LOMM

This appendix gives the formal LOMM specification behind the conceptual account in the main text. It is intended for readers who want to reimplement the generator or examine its assumptions and identification limits. Nothing in this appendix changes the reported results.

A.1 The Latent Transition

Equation 1 gives the continuous-time oscillator. The implementation advances its two-component state exactly over one observation interval using

$$\Phi_f(\Delta) = \exp(\mathbf{A}_f \Delta), \qquad \mathbf{A}_f = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix} \qquad \text{and} \qquad \Gamma_f(\Delta) = \int_0^\Delta \exp(\mathbf{A}_f u)\mathbf{b}\, du, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For the nonzero position-feedback values used in the simulation grids, the input map has the equivalent exact form

$$\Gamma_f(\Delta) = \mathbf{A}_f^{-1}\{\Phi_f(\Delta) - \mathbf{I}_2\}\mathbf{b}.$$

The integral definition remains valid when $\mathbf{A}_f$ is singular, including $\eta_f = 0$. If a root-based closed form is used numerically, a separate repeated-root branch is required when $\zeta_f^2 + 4\eta_f$ is close to zero.

The state components $x_{if,n}$ and $v_{if,n}$ are the level and its rate of change at t_n; η_f is position feedback, ζ_f is velocity feedback, and σ_{qf} is the standard deviation of the acceleration input. On the interval $[t_n, t_{n+1})$ the underlying continuous-time model is $\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}$ with $q_{if,n}$ held constant, so $\Phi_f(\Delta)$ propagates the deterministic state and $\Gamma_f(\Delta)$ loads the input. The inputs are independent across videos, dimensions, and transitions, and independent of initial states, activity schedules, and measurement errors. They are zero-order-hold acceleration inputs rather than increments of a continuous-time white-noise diffusion. The covariance added in one transition is $\Gamma_f(\Delta)\sigma_{qf}^2\Gamma_f(\Delta)^\top$.

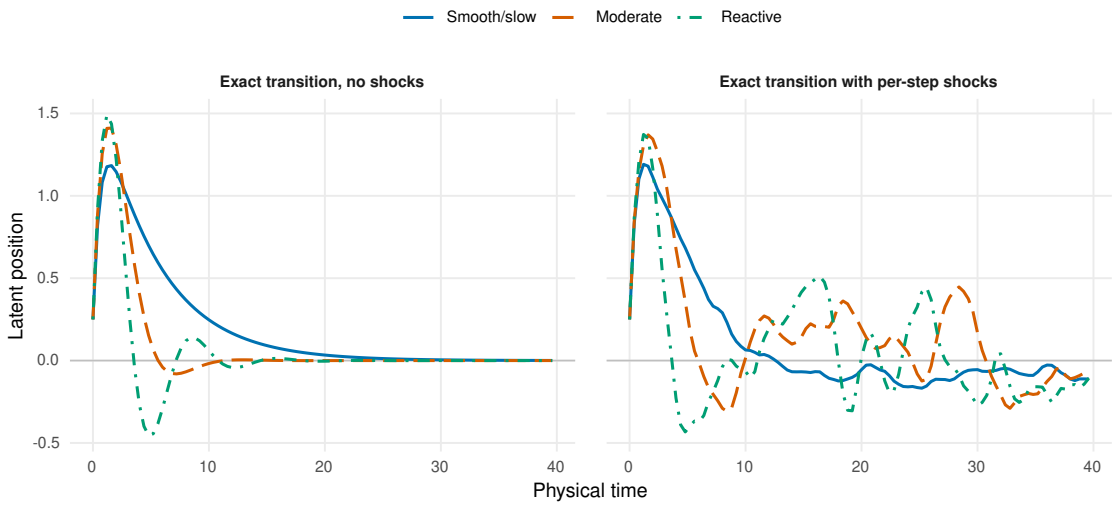


Figure A1: Illustrative latent trajectories for three uncoupled dimensions under the exact transition, shown without and with piecewise-constant acceleration inputs. The inputs create irregular peaks and reversals while the feedback parameters govern the underlying dynamic regime.

A.2 Dynamic Regimes and Finite-Horizon Amplification

The characteristic roots for dimension f are

$$r_{f,\pm} = \frac{\zeta_f \pm \sqrt{\zeta_f^2 + 4\eta_f}}{2}.$$

Continuous-time asymptotic stability requires $\eta_f < 0$ and $\zeta_f < 0$, and the roots are oscillatory when $\zeta_f^2 + 4\eta_f < 0$. The exact discrete transition is asymptotically stable when

$$\rho\{\Phi_f(\Delta)\} = \exp\left[\Delta \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}\right] < 1.$$

Negative η_f supplies restoring acceleration and negative ζ_f attenuates velocity, whereas positive ζ_f produces amplification. How much amplification actually appears depends on the observed horizon. For T observations, $H = (T-1)\Delta$. Let $\alpha(\mathbf{A}_f) = \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}$. The quantity

$$g_f(H) = \exp\{H\alpha(\mathbf{A}_f)\}$$

is a modal exponential-rate factor over that horizon. It does not by itself measure nonnormal transient amplification or the polynomial factor that can occur at a defective repeated root. A norm-based finite-

horizon state-amplification diagnostic is

$$G_f^{\max}(H) = \sup_{0 \le u \le H} \| \exp(\mathbf{A}_f u) \|_2,$$

and the corresponding level-only diagnostic is

$$G_{x,f}^{\max}(H) = \sup_{0 \le u \le H} \left\| \begin{bmatrix} 1 & 0 \end{bmatrix} \exp(\mathbf{A}_f u) \right\|_2 .$$

The Euclidean matrix norm is coordinate-dependent. Because the state combines a level and a rate of change, whose components have different units, this diagnostic is conditional on the chosen state coordinates and should be interpreted only after fixing a nondimensionalization or an explicit weighted state norm. A coordinate-aware definition is

$$G_{f,M}^{\max}(H) = \sup_{0 \le u \le H} \left\| \mathbf{M}^{1/2} \exp(\mathbf{A}_f u) \mathbf{M}^{-1/2} \right\|_2 ,$$

for a stated positive-definite state metric $\mathbf{M}$; the unweighted $G_f^{\max}(H)$ corresponds to $\mathbf{M} = \mathbf{I}_2$. Reporting the roots and $g_f(H)$ characterizes the modal regime over the observed horizon, but $g_f(H)$ should not be read as total finite-horizon amplification. Reporting the activity schedule separately keeps amplification in the latent dynamics distinct from nonstationarity created by the activity layer.

A.3 Initial States

The system is uncoupled: η_f, ζ_f, and σ_{qf} may vary by dimension, and off-diagonal dynamic effects are omitted. Study 2's primary simulation design initialized every dimension at $x_{if,1} = 0.25$ and $v_{if,1} = 2$. The sensitivity analysis used independent starting states. For a stable dimension, the stationary covariance $\mathbf{P}_f$ solves

$$\mathbf{P}_f = \Phi_f(\Delta) \mathbf{P}_f \Phi_f(\Delta)^\top + \Gamma_f(\Delta) \sigma_{qf}^2 \Gamma_f(\Delta)^\top,$$

whose solution is unique when $\rho\{\Phi_f(\Delta)\} < 1$. Drawing $\mathbf{s}_{if,1} \sim \mathcal{N}(\mathbf{0}, \mathbf{P}_f)$ makes the state process stationary before the activity layer is applied; time-varying activity modulation still renders $a_{if}(t) x_{if}(t)$ nonstationary. For a dimension with positive velocity feedback, the sensitivity draws $x_{if,1} \sim \text{Unif}(-0.25, 0.25)$ and $v_{if,1} \sim \text{Unif}(-2, 2)$. These independent starts remove the transient that all dimensions would otherwise share, while

leaving the observed horizon unchanged.

A.4 The Activity Layer at a Transition

Away from a discontinuity in the activity value, the ordinary product rule applies,

$$\frac{d}{dt}\{a_{if}(t)x_{if}(t)\} = a'_{if}(t)x_{if}(t) + a_{if}(t)x'_{if}(t).$$

At an abrupt transition, the derivative of the activity value contains an impulsive component in the distributional sense. Define $D_n x = x_{n+1} - x_n$. The exact discrete identity is

$$D_n(a_{if}x_{if}) = a_{if,n+1}D_n x_{if} + (D_n a_{if})x_{if,n},$$

where the first term uses the post-transition activity value and the second uses the pre-transition state. The corresponding finite-difference identity is

$$\frac{\tilde{x}_{if,n+1} - \tilde{x}_{if,n}}{\Delta} = a_{if,n+1}\frac{x_{if,n+1} - x_{if,n}}{\Delta} + \frac{a_{if,n+1} - a_{if,n}}{\Delta}x_{if,n}.$$

The second term is what a derivative-based method sees at a switch even when the oscillator itself has not accelerated, which is the reason for the gradual-transition sensitivity analysis. This convention attaches the new activity value to the post-transition observation, so the activity layer multiplies each completed latent state by the schedule value at the same observation index.

The activity schedule takes two nonnegative values, a_{high} and a_{low}. In non-switching families the dominant dimension remains at a_{high} for the whole record and the others remain at a_{low}. In switching families, dwell fractions determine how long each dimension remains dominant before the schedule reassigns a_{high}. Appendix B lists the numeric gains and dwell values used in the Study 2 families.

A.5 The Observation Layer

Let J denote the number of positive score channels, let $d = J - 1$, and let $\mathbf{V} \in \mathbb{R}^{J\times d}$ be a fixed orthonormal isometric log-ratio (ILR) basis satisfying $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_d$ and $\mathbf{V}^\top\mathbf{1}_J = \mathbf{0}$. The Study 1 basis used here is a normalized Helmert basis constructed in the stated 11-channel order. In the full observation model, $\mu \in \mathbb{R}^d$ is a fixed center, $\mathbf{b}_i \in \mathbb{R}^d$ is the clip baseline, $s_i > 0$ is the clip scale, $\kappa > 0$ is a global scale, and $\Lambda_i \in \mathbb{R}^{d\times m}$ maps the m activity-scaled latent positions into these coordinates. Conditional on the clip-specific terms

and Σ_i, measurement errors are independent over time and across videos, and independent of latent states, initial states, activity schedules, and acceleration inputs, with $\varepsilon_{in} \mid \Sigma_i \sim \mathcal{N}(\mathbf{0}, \Sigma_i)$. Dependence across coordinates follows Σ_i. Designs with fixed measurement parameters set $\Lambda_i = \Lambda$ and $\Sigma_i = \Sigma$ for every clip. The targeted residual-equicorrelation departure sets $\Sigma = s_\varepsilon^2\{(1-\rho)\mathbf{I}_p + \rho\mathbf{1}_p\mathbf{1}_p^\top\}$ with $\rho = 0.10$ across all p indicators, thereby introducing residual covariance both within and between the prespecified blocks. The clip scale multiplies both signal and residual, whereas the baseline is additive.

$$\mathbf{u}_{in} = \mu + \mathbf{b}_i + \kappa s_i \left(\Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in}\right).$$

The three baseline constructions differ only in where $\mathbf{b}_i$ is allowed to live. Study 1 draws it from a zero-mean Gaussian whose covariance is estimated from training clips in the full coordinate space. Study 2's between-clip-variability analysis instead uses

$$\mathbf{b}_i = \Lambda\beta_i, \qquad E(\beta_i) = \mathbf{0},$$

so that between-clip variation stays within the space spanned by the loadings and does not introduce structure outside the factor model. Study 2 conditions without baseline and scale differences fix $\mathbf{b}_i = \mathbf{0}$ and $s_i = 1$, giving

$$\mathbf{u}_{in} = \mu + \kappa \left(\Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in}\right).$$

In the matched-method and extended factorial designs, Λ_i and the diagonal of Σ_i receive small clip-specific perturbations to the loadings and residual variances. The targeted sensitivity and between-clip-variability designs use fixed equal values except where the stated condition changes them. Thus $\mathbf{b}_i = \mathbf{0}$ and $s_i = 1$ remove baseline and overall-scale differences; they do not by themselves imply a common loading matrix or residual covariance.

The synthetic continuous indicators used in Study 2's continuous-indicator recovery designs are the continuous-indicator vector before any bounded-score mapping,

$$\mathbf{z}_{in} = \Lambda_i \tilde{\mathbf{x}}_{in} + \varepsilon_{in},$$

in which each of the $p = mk$ indicators loads on one dimension and has diagonal measurement error.

For bounded output, LOMM maps the coordinates to the open probability simplex. The centered log-ratio (CLR) expresses each channel relative to the geometric mean of all channels. The isometric log-ratio (ILR) represents the same information in $d = J - 1$ orthonormal coordinates. Formally,

$$\mathrm{clr}(\mathbf{p}) = \log \mathbf{p} - J^{-1}\{\mathbf{1}_J^\top \log(\mathbf{p})\}\mathbf{1}_J, \qquad \mathrm{ilr}_{\mathbf{V}}(\mathbf{p}) = \mathbf{V}^\top \mathrm{clr}(\mathbf{p}),$$

with the logarithm applied componentwise. For the canonical bounded LOMM output, applying softmax to $\mathbf{V}\mathbf{u}_{in}/\tau$ gives

$$\mathrm{ilr}_{\mathbf{V}}(\mathbf{p}_{in}) = \frac{\mathbf{u}_{in}}{\tau}, \qquad \mathrm{clr}(\mathbf{p}_{in}) = \frac{\mathbf{V}\mathbf{u}_{in}}{\tau},$$

the second because $\mathbf{V}\mathbf{u}_{in}$ lies in the zero-sum log-ratio subspace. The fixed basis excludes the unidentified common-logit direction and the softmax produces strictly positive rows summing to one, so the scores identify the scaled coordinates $\mathbf{u}_{in}/\tau$ exactly. A common-logit component remains unobservable, and the unscaled coordinates and τ are not separately identified without an external scale constraint.

The targeted Study 2 bounded-score departure uses the $p = mk$ continuous-indicator vector directly. With $J = p$, it computes

$$\mathbf{p}_{in}^{\mathrm{dep}} = \mathrm{softmax}\left(\frac{\mathbf{z}_{in}}{\tau}\right), \qquad \mathrm{clr}\left(\mathbf{p}_{in}^{\mathrm{dep}}\right) = \frac{1}{\tau}\mathbf{P}_p\mathbf{z}_{in}, \qquad \mathbf{P}_p = \mathbf{I}_p - \frac{\mathbf{1}_p\mathbf{1}_p^\top}{p}.$$

For any orthonormal ILR basis $\mathbf{V}_p \in \mathbb{R}^{p\times(p-1)}$, $\mathbf{V}_p\mathbf{V}_p^\top = \mathbf{P}_p$. Defining $\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{V}_p^\top\mathbf{z}_{in}$ therefore gives $\mathbf{V}_p\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{P}_p\mathbf{z}_{in}$. Softmax is invariant to the removed row mean, so the direct-softmax branch is equivalent to applying the basis form to $\mathbf{u}_{in}^{\mathrm{dep}}$. The executed analysis used the p raw scores and the p CLR components, which satisfy a zero-sum constraint; it did not analyze the $(p-1)$ ILR coordinates. The common-logit direction in $\mathbf{z}_{in}$ is unidentifiable after this transformation.

A.6 Converting Period and Damping to Feedback Parameters

Study 1's search was parameterized by natural period and damping ratio rather than by η_f and ζ_f directly. If P_f is the natural period in observation intervals and ξ_f is the damping ratio, then

$$\omega_{n,f} = \frac{2\pi}{P_f\Delta}, \qquad \eta_f = -\omega_{n,f}^2, \qquad \zeta_f = -2\xi_f\omega_{n,f},$$

and the damped angular frequency is $\omega_{d,f} = \omega_{n,f}\sqrt{1-\xi_f^2}$. With $P_f = 12$, $\xi_f = 0.5$, and $\Delta = 1$, these give $\eta_f = -0.2742$ and $\zeta_f = -0.5236$; scaling the inputs to unit stationary position variance gives $\sigma_{qf} = 0.5418$.

A.7 Identifiability

For any nonsingular diagonal matrix $\mathbf{D}$,

$$\Lambda\tilde{\mathbf{x}}_{in} = (\Lambda\mathbf{D}^{-1})(\mathbf{D}\tilde{\mathbf{x}}_{in}),$$

so loading magnitude and latent-trajectory scale are not jointly identified without an explicit scale constraint. Activity gain and acceleration-input scale contribute to latent amplitude in the same way. The softmax layer adds invariance to a common location shift and a tradeoff between the temperature and the coordinate scale. Dimension labels are recoverable only up to permutation.

Sampling cadence adds an aliasing restriction. For an underdamped dimension, write

$$r_{f,\pm} = \frac{\zeta_f}{2} \pm i\omega_f, \qquad \omega_f = \frac{1}{2}\sqrt{-(\zeta_f^2 + 4\eta_f)}.$$

At observation times $t_n = (n-1)\Delta$, the frequencies ω_f and $\omega_f + 2\pi q/\Delta$ for integer q generate the same sampled sine and cosine sequences after an appropriate change in initial velocity. Identifying continuous-time oscillator parameters from data therefore requires a cadence-dependent restriction such as

$$0 < \omega_f < \frac{\pi}{\Delta},$$

together with a specified acceleration-input model and constraints on the latent state basis. Estimating LOMM from data would in total require sign and scale constraints on the dimensions, normalization of the activity schedule, anchoring of either the loading scale or the input scale, a reference-logit or sum-to-zero constraint, a fixed or otherwise identified temperature, and this frequency restriction. Every oscillatory mode used here satisfies the restriction: Study 1 uses $\Delta = 1$, giving the most restrictive bound $\pi \approx 3.14$ against a largest angular frequency of approximately 0.78, and Study 2 uses $\Delta \in \{0.25, 0.4\}$, giving bounds of approximately 12.57 and 7.85 against a largest angular frequency of approximately 1.04. The simulations do not estimate any of these parameters.

Appendix B: Study Designs and Implementation Details

This appendix collects the design and implementation details moved from the main text. Study 1 evaluates resemblance to real bounded-score series. Study 2 evaluates recovery of a prespecified factor structure from synthetic continuous indicators. These targets remain distinct.

B.1 Study 1 Scoring and Calibration

The Study 1 sample was assembled one clip at a time, with MAFW category labels used to diversify the sample rather than as validation targets. The main analysis contains 3,557 observations in 188 uninterrupted segments from 100 clips. Segment lengths range from 16 to 50 observations, with a median of 35. The gap-interpolated sensitivity analysis contains 3,632 observations in 129 segments, with lengths from 17 to 50 and a median of 36.5.

transforEmotion sampled up to 50 frames uniformly from each video and used the OpenAI CLIP ViT-Large/14 model family. A Haar cascade detected faces, and the largest detected face was expanded by 50 pixels on each side before classification. Frames without a detected face received missing scores. Eleven lowercase class labels were passed directly to the text encoder without a sentence template: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. Burned-in subtitles were not masked and could contribute when they fell inside the padded face crop.

Scores were required to lie in $[0, 1]$. A complete row was retained when its sum was positive, finite, and within 10^{-6} of one, then divided by that sum in the fixed channel order. Values were not truncated and zero replacement was disabled. The main analysis retained complete rows. The sensitivity analysis interpolated internal gaps of at most two frames within a segment. The scoring-time package release and exact checkpoint revision were not archived, so exact frame rescoring cannot be guaranteed; the retained scores support the reported analyses.

The comparison used eight statistics. Median level, interquartile range of levels, interquartile range of observation-to-observation changes, lag-1 autocorrelation, entropy of monotonic run lengths, and normalized spectral entropy were calculated for each channel. Effective rank and total variance described all channels jointly. Each statistic was evaluated on both bounded scores and log-ratio coordinates.

For every statistic and representation, disagreement between simulated and real clip distributions was measured by the absolute difference between medians and the one-dimensional Wasserstein distance. Both were standardized by the spread of the real values, using the interquartile range when available, followed by 1.4826 times the median absolute deviation, the standard deviation, or one. The larger standardized distance was

Table B1: The six Study 2 scenario families. Dwell values are fractions of the record length.

Code	Family	Switching	Active / background gain	Dwell fraction	Dynamic rule
A	Weak expression, one dominant dimension	No	0.35 / 0.05	1.00 / 1.00	Stable, low input scale
B	Strong expression, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Stable, standard input scale
C	Strong expression, switching dimension	Yes	1.00 / 0.08	0.20 / 0.45	Stable, standard input scale
D	Strong expression, rapid switching	Yes	1.25 / 0.08	0.03 / 0.10	Stable, high input scale
E	Amplifying trajectories, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Positive velocity feedback
F	Amplifying trajectories, switching dimension	Yes	1.10 / 0.10	0.12 / 0.30	Positive velocity feedback

retained. The composite averaged the eight statistics within each representation and weighted the two representations equally.

Clips were divided into five deterministic folds within MAFW category and length order, leaving 20 test clips per fold. The coordinate center, loading directions, loading scales, and covariance of clip baselines were estimated from training clips only. The first three eigenvectors of the within-clip covariance supplied the loading directions. Their orientation and scale were fixed deterministically. The baseline covariance was shrunk halfway toward its diagonal and eigenvalues were floored at 10^{-8}.

Within each fold and configuration, the search crossed natural periods $\{8, 12, 24\}$, damping ratios $\{0.05, 0.20, 0.50\}$, and observation scales $\kappa \in \{0.75, 1.00, 1.25\}$. Three paired training datasets evaluated each of the 27 candidates. Selection minimized the equally weighted representation-level discrepancy, with deterministic tie-breaking. All folds selected a 12-observation period, damping ratio 0.50, and $\kappa = 1$ for the baseline-and-scale configuration.

Each selected configuration generated 100 datasets per test fold, matching the held-out observation indices and segment structure. The comparison concerns distributions of summary statistics across clips, rather than frame-level reconstruction. Uncertainty used 2,000 paired bootstrap resamples of the 20 clips within each fold. The eight configurations are the factorial combinations of clip baseline, clip scale, and brief activity transition.

B.2 Study 2 Simulation Designs

For each clip, LOMM sets or draws the starting state and acceleration inputs according to the configuration, advances each latent dimension, applies the activity schedule, and constructs the observed indicators. The primary, shorter-series, and extended factorial designs use synthetic continuous indicators. Clips share the block partition and nominal measurement settings, with clip-specific loading and uniqueness jitter. The targeted conditions use fixed equal measurement parameters except for the stated departure.

Table B2: Targeted departures from the fixed equal-measurement condition.

Departure	Specification
Independent starting states	Stationary zero-mean starts for stable dimensions; bounded independent starts for amplifying dimensions
Gradual activity transitions	A five-observation transition replaces each abrupt switch
Cross-loadings	Small normal cross-loadings, truncated to [-0.10, 0.10]
Residual equicorrelation	Correlation 0.10 between every residual pair, including pairs from different dimensions
Correlated inputs	Acceleration-input correlation 0.15 between dimensions
Bounded scores	Direct softmax of the p indicators with tau = 1, analyzed as raw scores or CLR components

In non-switching scenarios, one dimension remains dominant for a whole clip and the dominant dimension rotates across clips. Switching scenarios sample dwell intervals and change the dominant dimension over time. Families A through D use stable dynamics. Families E and F use positive velocity feedback and serve as finite-horizon amplification stress tests.

The primary $T = 100$ design contains 21,600 datasets. It crosses six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ clips, loadings $\{0.40, 0.60, 0.80\}$, residual standard deviations $\{0.125, 0.25, 0.50\}$, four indicators per dimension, and 100 paired replications. The shorter-series design repeats the same 21,600 datasets with $T = 25$, $\Delta = 0.4$, and duration 9.6. Every method receives the same dataset within each condition and replication.

The targeted analyses fix $N = 100$, $T = 100$, loading 0.60, residual standard deviation 0.25, and four indicators per dimension. They cross all six families, two dimensionalities, and 50 paired replications while changing one feature at a time.

The between-clip analysis uses 80 paired datasets from stable families A through D, $m \in \{2, 3\}$, and 10 replications. Each dataset has $N = 100$, $T = 100$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and independent stationary starts. Its baseline varies within the loading space, and its scale varies across clips.

The extended factorial simulation evaluates DynEGA on 518,400 datasets. It crosses $N \in \{50, 100\}$, $T \in \{100, 300\}$, $m \in \{2, 3, 4\}$, $k \in \{4, 6\}$, three loading levels, three residual-error levels, six scenarios, and 400 replications. The $T = 100$ conditions use $\Delta = 0.4$ and the $T = 300$ conditions use $\Delta = 0.25$ while holding per-step input variance fixed. Their difference therefore combines record length, observation interval, physical duration, and discrete input specification.

The public lomm_observe() function documents the canonical observation step and returns the selected profile, implementation hashes, seed states, and loading and basis hashes. Study 2 uses the benchmark generator documented in the analysis code, which returns continuous indicators for the primary designs and applies the direct-softmax and CLR transformation for the targeted bounded-score departure.

B.3 Method Implementations

DynEGA estimates first derivatives from time-delay embeddings, constructs an EBICglasso partial-correlation network, and groups its positive edges (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017). Video identity prevents embedding windows from crossing clip boundaries. An explicit search selects the embedding dimension by the Total Entropy Fit Index, with shorter embeddings breaking ties. Candidate dimensions are $\{5, 10, 15, 25, 50\}$ for $T = 100$ and $\{5, 6, 10, 12\}$ for $T = 25$. EGAnet 2.3.0 uses tau 1, delta 1, first derivatives, automatic correlations, pairwise missing-data handling, and the glasso model. The derivative scale does not affect the correlation matrix because conversion from observation units to physical time multiplies every derivative by the same positive constant.

Static EGA applies EBICglasso to the contemporaneous correlation matrix of the stacked indicator levels and then uses the same network-processing and community-detection rules as DynEGA. It ignores the observation order. This makes it a time-agnostic benchmark for the grouping target and a computational benchmark for the temporal procedures.

GraphicalVAR uses mlGraphicalVAR with clip identity as the grouping variable, within-clip centering, variable standardization, and gamma 0.5 (Epskamp et al., 2018; Epskamp, 2024). Lagged pairs never cross clip boundaries. The grouping uses the returned fixedPCC contemporaneous residual partial-correlation matrix. The diagonal is removed and the matrix is treated as an undirected weighted graph.

GIMME fits each standardized clip separately with unified structural equation modeling and vector autoregression (Gates & Molenaar, 2012; Lane et al., 2019). Standardized residual covariance matrices from successful fits are aligned by variable name, averaged across clips, and symmetrized. The resulting graph is a residual-covariance graph. Unequal residual variances can therefore influence its weights. This conversion was built for the present comparison and differs from typical idiographic use.

For every method, negative graph weights are set to zero. If no positive edge remains, absolute values of the original off-diagonal weights are used; only an all-zero matrix produces singleton communities. The integrated outputs do not record the frequency of this fallback. Louvain at resolution 1.0 is primary (Blondel et al., 2008). Leiden and Walktrap are applied to the same fitted network as sensitivity analyses (Pons & Latapy, 2005; Traag et al., 2019). A deterministic offset from the dataset seed initializes each method fit.

GraphicalVAR and GIMME were omitted from the 518,400-dataset extended factorial simulation because the full design was computationally impractical. Both were included in the primary and shorter-series designs. GraphicalVAR was also included in the targeted and between-clip analyses. GIMME was included in the between-clip analysis and excluded from the targeted one-at-a-time departures.

B.4 Outcomes, Uncertainty, and Computing

The valid-output rate records whether a procedure returned a network that could be grouped. The conditional correct-dimension rate records correct dimensionality among valid outputs. The headline overall correct-dimension recovery rate requires both events, so errors and timeouts remain in its denominator.

Adjusted Rand index and variation of information compare the full estimated and prespecified partitions among valid outputs. Assignment accuracy matches estimated groups to dimensions to maximize correct indicator assignments. Every indicator remains in the denominator, and indicators in unmatched groups count as incorrect.

Trajectory-summary diagnostics are constructed after grouping. Indicators are standardized and each estimated group is summarized by its first principal component, with a row mean used when a component cannot be computed. Correlation pairs each prespecified dimension with the estimated summary having the largest absolute correlation. NRMSE uses one-to-one matching after sign reversal when needed. An unmatched data-generating dimension receives the penalty 2. All-attempt failure-adjusted versions assign correlation 0 and NRMSE 2 when no valid network output exists.

Uncertainty uses 2,000 bootstrap resamples of whole datasets, stratified by condition. Monte Carlo standard errors use delete-one-dataset jackknifing. The three community-detection algorithms reuse the same fitted structure and therefore assess the grouping step rather than estimator variability.

Analyses used R 4.5.1 on Rocky Linux 8.8 with dual Intel Xeon Silver 4310 processors, 48 logical threads, and 125 GiB RAM. Key versions were EGAnet 2.3.0, graphicalVAR 0.3.4, gimme 10.0, igraph 2.2.1, qgraph 1.9.8, psych 2.5.6, lavaan 0.6-20, dplyr 1.1.4, and ggplot2 4.0.1. GraphicalVAR and GIMME used a 900-second limit per fit in the primary and shorter designs.

Runtime is elapsed wall-clock time for attempts returning a valid network output. It includes structure estimation, network extraction, and community detection. One fitted structure is reused for Louvain, Leiden, and Walktrap, so feasibility counts unique fits and accuracy counts fit-and-algorithm results. Runtime values are specific to this implementation and machine.

Appendix C: Supplementary Results

C.1 Full Results of the Simulated-versus-Real Comparison

The complete comparison of all eight configurations is reported below for both analyses. The tables retain the bootstrap means, estimated biases, basic reverse-percentile intervals, and percentile intervals so that the displacement can be inspected directly. These quantities diagnose the behavior of the clip-resampling distribution; they are not treated as uncertainty intervals around the absolute plug-in profile discrepancies. The following table then separates the empirical-baseline plus clip-scale profile into its eight component metrics. This second table preserves discrepancies that are obscured by the composite mean, including the remaining differences in amplitude dispersion and first-difference variability.

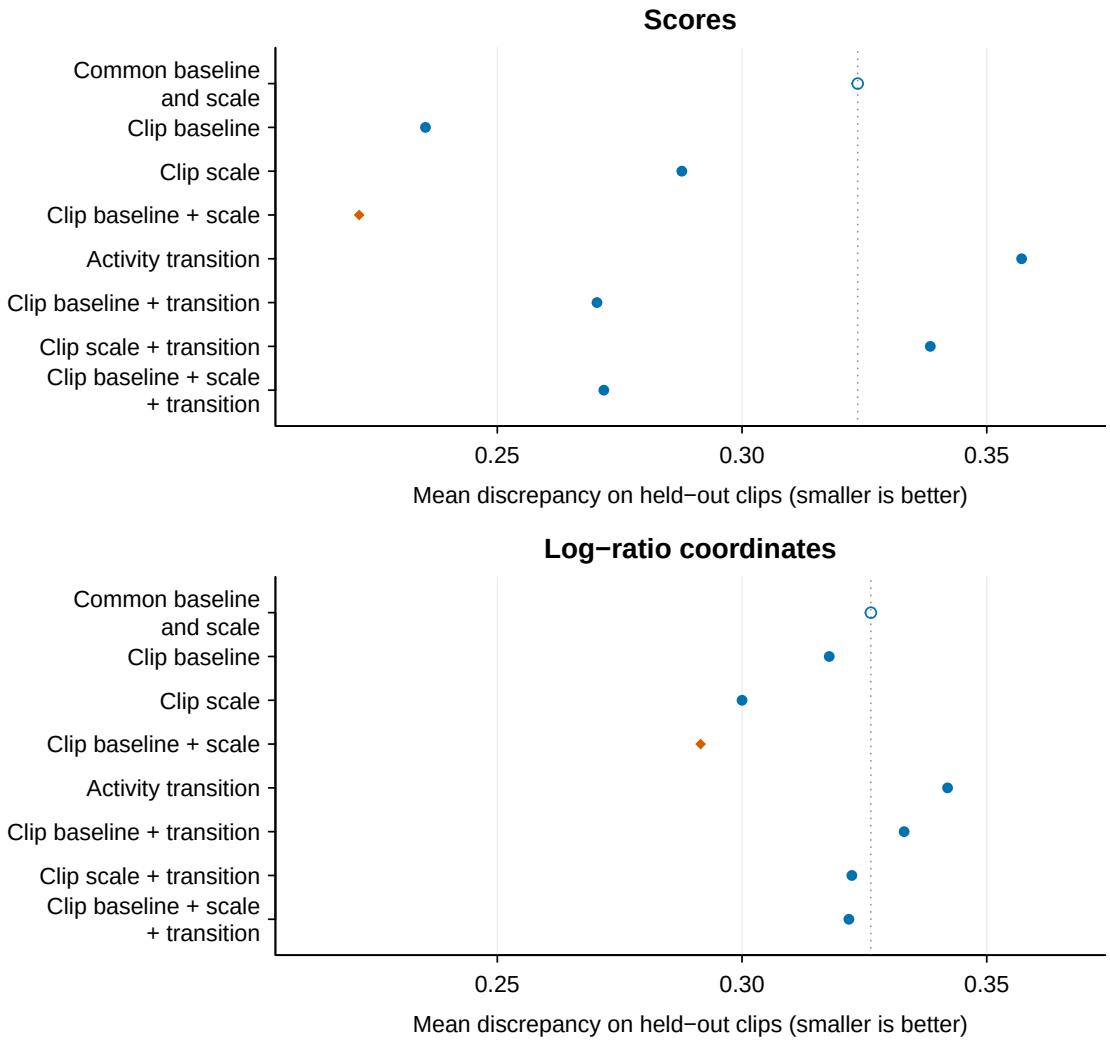


Figure C1: Discrepancy between simulated and real series for all eight configurations, shown for bounded scores and log-ratio coordinates. Smaller values indicate closer agreement. The main text shows the score panel only; configuration selection weighted both representations equally.

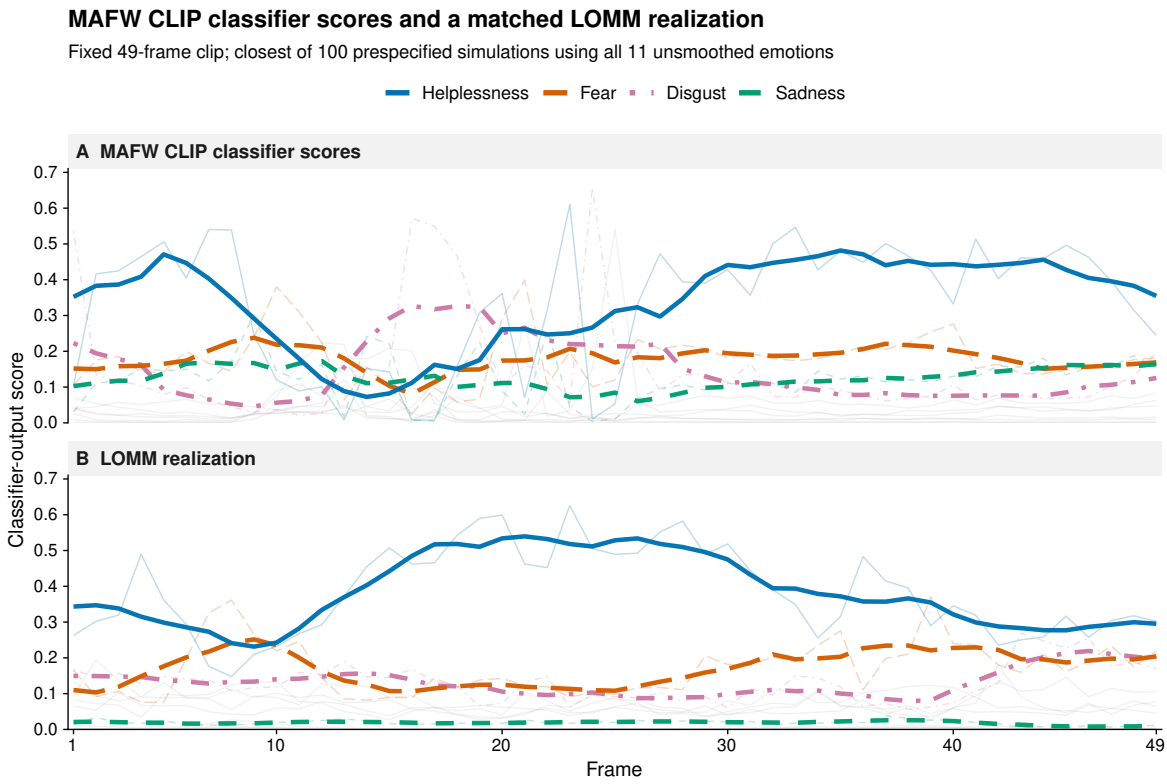


Figure C2: Real MAFW classifier scores and the closest of 100 LOMM realizations generated for the same 49-observation clip with the baseline-and-scale configuration. The selected realization had a score RMSE of 0.105 and an observation-to-observation-change RMSE of 0.083. Selection after generation makes this a best-case illustration; the cross-fitted comparison carries the aggregate evidence.

Table C1: Discrepancy between simulated and real score series for the eight configurations, complete frames analysis. Each estimate averages eight summary statistics within a representation across five equally weighted folds, 100 held-out video clips, and 100 simulated datasets per fold and configuration. Smaller values indicate closer agreement with the real scores. Bootstrap bias is the bootstrap mean minus the estimate. Basic reverse-percentile and percentile intervals are reported as diagnostics rather than as uncertainty intervals for individual configurations. They are conditional on the settings selected within each training fold.

Representation	Configuration	Mean	MCSE	Bootstrap mean	Bias	Basic 95% interval	Percentile 95% interval
Scores	Common baseline-and-scale configuration	0.3236	0.0007	0.3455	0.0219	[0.2691, 0.3322]	[0.3151, 0.3782]
Scores	Clip baseline	0.2353	0.0009	0.2723	0.0370	[0.1686, 0.2268]	[0.2439, 0.3020]
Scores	Clip scale	0.2877	0.0010	0.3233	0.0356	[0.2197, 0.2803]	[0.2951, 0.3557]
Scores	Clip baseline + scale	0.2218	0.0013	0.2701	0.0484	[0.1433, 0.1992]	[0.2444, 0.3002]
Scores	Activity transition	0.3571	0.0012	0.3830	0.0259	[0.2898, 0.3674]	[0.3468, 0.4245]
Scores	Clip baseline + activity transition	0.2704	0.0013	0.3090	0.0387	[0.1958, 0.2627]	[0.2780, 0.3449]
Scores	Clip scale + activity transition	0.3385	0.0013	0.3708	0.0324	[0.2641, 0.3424]	[0.3345, 0.4128]
Scores	Clip baseline + scale + activity transition	0.2718	0.0014	0.3149	0.0431	[0.1891, 0.2618]	[0.2817, 0.3545]
Log ratio	Common baseline-and-scale configuration	0.3263	0.0011	0.3538	0.0275	[0.2582, 0.3346]	[0.3180, 0.3944]
Log ratio	Clip baseline	0.3178	0.0011	0.3476	0.0298	[0.2478, 0.3241]	[0.3115, 0.3878]
Log ratio	Clip scale	0.3000	0.0023	0.3453	0.0453	[0.2180, 0.2851]	[0.3149, 0.3821]
Log ratio	Clip baseline + scale	0.2915	0.0023	0.3392	0.0476	[0.2065, 0.2744]	[0.3087, 0.3766]
Log ratio	Activity transition	0.3420	0.0013	0.3778	0.0358	[0.2604, 0.3447]	[0.3393, 0.4236]
Log ratio	Clip baseline + activity transition	0.3331	0.0013	0.3713	0.0382	[0.2490, 0.3327]	[0.3335, 0.4172]
Log ratio	Clip scale + activity transition	0.3224	0.0018	0.3695	0.0470	[0.2288, 0.3134]	[0.3315, 0.4160]
Log ratio	Clip baseline + scale + activity transition	0.3218	0.0019	0.3699	0.0481	[0.2294, 0.3104]	[0.3332, 0.4143]

Table C2: Discrepancy between simulated and real score series for the eight configurations, gap interpolated analysis. Each estimate averages eight summary statistics within a representation across five equally weighted folds, 100 held-out video clips, and 100 simulated datasets per fold and configuration. Smaller values indicate closer agreement with the real scores. Bootstrap bias is the bootstrap mean minus the estimate. Basic reverse-percentile and percentile intervals are reported as diagnostics rather than as uncertainty intervals for individual configurations. They are conditional on the settings selected within each training fold.

Representation	Configuration	Mean	MCSE	Bootstrap mean	Bias	Basic 95% interval	Percentile 95% interval
Scores	Common baseline-and-scale configuration	0.3164	0.0007	0.3389	0.0226	[0.2614, 0.3231]	[0.3097, 0.3714]
Scores	Clip baseline	0.2290	0.0009	0.2660	0.0369	[0.1618, 0.2183]	[0.2398, 0.2963]
Scores	Clip scale	0.2803	0.0010	0.3168	0.0366	[0.2103, 0.2713]	[0.2892, 0.3502]
Scores	Clip baseline + scale	0.2152	0.0012	0.2639	0.0487	[0.1366, 0.1909]	[0.2395, 0.2939]
Scores	Activity transition	0.3647	0.0012	0.3886	0.0240	[0.2996, 0.3775]	[0.3518, 0.4297]
Scores	Clip baseline + activity transition	0.2797	0.0014	0.3151	0.0353	[0.2075, 0.2763]	[0.2832, 0.3520]
Scores	Clip scale + activity transition	0.3461	0.0014	0.3768	0.0307	[0.2733, 0.3526]	[0.3395, 0.4188]
Scores	Clip baseline + scale + activity transition	0.2819	0.0015	0.3222	0.0403	[0.2016, 0.2752]	[0.2887, 0.3623]
Log ratio	Common baseline-and-scale configuration	0.3209	0.0011	0.3485	0.0276	[0.2531, 0.3296]	[0.3123, 0.3888]
Log ratio	Clip baseline	0.3124	0.0011	0.3424	0.0300	[0.2433, 0.3188]	[0.3061, 0.3815]
Log ratio	Clip scale	0.2954	0.0023	0.3404	0.0450	[0.2145, 0.2819]	[0.3089, 0.3763]
Log ratio	Clip baseline + scale	0.2869	0.0023	0.3342	0.0474	[0.2036, 0.2710]	[0.3027, 0.3701]
Log ratio	Activity transition	0.3528	0.0015	0.3864	0.0335	[0.2725, 0.3585]	[0.3471, 0.4331]
Log ratio	Clip baseline + activity transition	0.3439	0.0015	0.3799	0.0360	[0.2617, 0.3466]	[0.3413, 0.4261]
Log ratio	Clip scale + activity transition	0.3337	0.0019	0.3782	0.0444	[0.2418, 0.3280]	[0.3395, 0.4257]
Log ratio	Clip baseline + scale + activity transition	0.3360	0.0020	0.3812	0.0452	[0.2432, 0.3288]	[0.3431, 0.4287]

Table C3: Discrepancy for each summary statistic under the baseline-and-scale configuration, complete frames analysis. Results are shown for both representations; smaller values indicate closer agreement with the real scores. For the six single-channel statistics, a unit is one clip-channel summary; for effective rank and total variance, a unit is one clip-level summary. Unit columns report the range across folds and simulation repetitions. The bootstrap means, bias estimates, and 95% intervals are diagnostics rather than uncertainty intervals for the individual discrepancies. They are conditional on the settings selected within each fold.

Representation	Metric	Mean	MCSE	Bias	Basic 95% interval	Percentile 95% interval	Empirical units/fold	Simulated units/fold
Scores	Level median	0.1434	0.0020	0.0379	[0.0657, 0.1364]	[0.1503, 0.2210]	220	220
Scores	Level IQR	0.1456	0.0020	0.0387	[0.0748, 0.1331]	[0.1581, 0.2163]	220	220
Scores	First-difference IQR	0.1846	0.0024	0.0244	[0.1235, 0.1923]	[0.1768, 0.2456]	220	220
Scores	Lag-1 autocorrelation	0.2143	0.0016	0.0241	[0.1221, 0.2419]	[0.1867, 0.3064]	220	220
Scores	Run-length entropy	0.1160	0.0018	0.0396	[0.0467, 0.1029]	[0.1290, 0.1852]	220	220
Scores	Spectral entropy	0.2978	0.0031	0.0139	[0.2122, 0.3513]	[0.2444, 0.3835]	209–220	209–220
Scores	Effective rank	0.3227	0.0033	0.0745	[0.1595, 0.3252]	[0.3202, 0.4859]	20	20
Scores	Total variance	0.3499	0.0058	0.1337	[0.0726, 0.3002]	[0.3995, 0.6271]	20	20
Log ratio	Level median	0.0701	0.0008	0.0250	[0.0322, 0.0559]	[0.0843, 0.1081]	220	220
Log ratio	Level IQR	0.2969	0.0054	0.0497	[0.1777, 0.3025]	[0.2913, 0.4161]	220	220
Log ratio	First-difference IQR	0.4423	0.0073	0.0340	[0.3177, 0.4903]	[0.3943, 0.5669]	220	220
Log ratio	Lag-1 autocorrelation	0.2903	0.0021	0.0206	[0.1778, 0.3465]	[0.2342, 0.4028]	220	220
Log ratio	Run-length entropy	0.1323	0.0020	0.0343	[0.0608, 0.1305]	[0.1340, 0.2037]	220	220
Log ratio	Spectral entropy	0.3979	0.0035	0.0083	[0.3031, 0.4734]	[0.3224, 0.4927]	209–220	209–220
Log ratio	Effective rank	0.3455	0.0024	0.0588	[0.1977, 0.3672]	[0.3237, 0.4932]	20	20
Log ratio	Total variance	0.3570	0.0068	0.1505	[0.1074, 0.2706]	[0.4435, 0.6067]	20	20

Table C4: Discrepancy for each summary statistic under the baseline-and-scale configuration, gap interpolated analysis. Results are shown for both representations; smaller values indicate closer agreement with the real scores. For the six single-channel statistics, a unit is one clip-channel summary; for effective rank and total variance, a unit is one clip-level summary. Unit columns report the range across folds and simulation repetitions. The bootstrap means, bias estimates, and 95% intervals are diagnostics rather than uncertainty intervals for the individual discrepancies. They are conditional on the settings selected within each fold.

Representation	Metric	Mean	MCSE	Bias	Basic 95% interval	Percentile 95% interval	Empirical units/fold	Simulated units/fold
Scores	Level median	0.1431	0.0020	0.0377	[0.0651, 0.1366]	[0.1495, 0.2210]	220	220
Scores	Level IQR	0.1417	0.0020	0.0397	[0.0713, 0.1281]	[0.1554, 0.2122]	220	220
Scores	First-difference IQR	0.1879	0.0025	0.0232	[0.1291, 0.1966]	[0.1792, 0.2467]	220	220
Scores	Lag-1 autocorrelation	0.1937	0.0016	0.0236	[0.1061, 0.2183]	[0.1691, 0.2813]	220	220
Scores	Run-length entropy	0.1179	0.0018	0.0354	[0.0509, 0.1096]	[0.1262, 0.1849]	220	220
Scores	Spectral entropy	0.2724	0.0027	0.0187	[0.1905, 0.3075]	[0.2374, 0.3543]	220	220
Scores	Effective rank	0.3225	0.0033	0.0747	[0.1601, 0.3244]	[0.3205, 0.4848]	20	20
Scores	Total variance	0.3426	0.0058	0.1362	[0.0593, 0.2877]	[0.3975, 0.6259]	20	20
Log ratio	Level median	0.0687	0.0008	0.0258	[0.0301, 0.0536]	[0.0839, 0.1073]	220	220
Log ratio	Level IQR	0.3032	0.0054	0.0462	[0.1867, 0.3120]	[0.2943, 0.4197]	220	220
Log ratio	First-difference IQR	0.4639	0.0075	0.0321	[0.3462, 0.5079]	[0.4199, 0.5816]	220	220
Log ratio	Lag-1 autocorrelation	0.2645	0.0019	0.0234	[0.1505, 0.3161]	[0.2129, 0.3786]	220	220
Log ratio	Run-length entropy	0.1280	0.0021	0.0334	[0.0528, 0.1272]	[0.1288, 0.2032]	220	220
Log ratio	Spectral entropy	0.3674	0.0033	0.0095	[0.2796, 0.4355]	[0.2993, 0.4553]	220	220
Log ratio	Effective rank	0.3452	0.0024	0.0584	[0.1971, 0.3662]	[0.3242, 0.4932]	20	20
Log ratio	Total variance	0.3541	0.0068	0.1502	[0.1100, 0.2658]	[0.4424, 0.5982]	20	20

C.2 Grouping and Trajectory Diagnostics

This appendix reports two types of post-estimation diagnostic. Grouping diagnostics evaluate recovered indicator groupings among valid network outputs. Trajectory-summary diagnostics compare summaries constructed from recovered groups with the activity-scaled latent trajectories; the trajectories are not direct latent-state estimates produced by the methods.

Table C5: Louvain grouping diagnostics for the matched-method simulation with $T = 100$. Each diagnostic uses the datasets satisfying its stated conditioning rule. Monte Carlo standard errors are computed across simulated datasets, and 95% intervals resample whole datasets, stratified by condition.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid outputs	Correct-dimensional outputs
DynEGA	Signed dimension error	0.166	0.0057	[0.154, 0.177]	21600	21600	20282
DynEGA	Absolute dimension error	0.168	0.0057	[0.157, 0.180]	21600	21600	20282
DynEGA	Adjusted Rand index	0.935	0.0015	[0.932, 0.938]	21600	21600	20282
DynEGA	Variation of information	0.095	0.0023	[0.091, 0.100]	21600	21600	20282
DynEGA	Assignment accuracy given valid network output	0.965	0.0009	[0.963, 0.966]	21600	21600	20282
GraphicalVAR	Signed dimension error	0.979	0.0171	[0.947, 1.011]	20329	20329	16784
GraphicalVAR	Absolute dimension error	0.979	0.0171	[0.945, 1.012]	20329	20329	16784
GraphicalVAR	Adjusted Rand index	0.867	0.0022	[0.863, 0.872]	20329	20329	16784
GraphicalVAR	Variation of information	0.183	0.0031	[0.177, 0.189]	20329	20329	16784
GraphicalVAR	Assignment accuracy given valid network output	0.902	0.0016	[0.899, 0.906]	20329	20329	16784
GIMME	Signed dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Absolute dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Adjusted Rand index	0.733	0.0057	[0.722, 0.744]	4846	4846	3795
GIMME	Variation of information	0.337	0.0073	[0.323, 0.351]	4846	4846	3795
GIMME	Assignment accuracy given valid network output	0.877	0.0027	[0.872, 0.882]	4846	4846	3795
Static EGA	Signed dimension error	0.799	0.0159	[0.770, 0.829]	21526	21526	19093
Static EGA	Absolute dimension error	0.799	0.0159	[0.767, 0.829]	21526	21526	19093
Static EGA	Adjusted Rand index	0.894	0.0021	[0.889, 0.898]	21526	21526	19093
Static EGA	Variation of information	0.147	0.0029	[0.142, 0.153]	21526	21526	19093
Static EGA	Assignment accuracy given valid network output	0.922	0.0015	[0.919, 0.925]	21526	21526	19093

Table C6: Louvain grouping diagnostics for the matched-method simulation with $T = 25$. Each diagnostic uses the datasets satisfying its stated conditioning rule. Monte Carlo standard errors are computed across simulated datasets, and 95% intervals resample whole datasets, stratified by condition.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid outputs	Correct-dimensional outputs
DynEGA	Signed dimension error	0.492	0.0102	[0.471, 0.512]	21600	21600	18636
DynEGA	Absolute dimension error	0.494	0.0102	[0.474, 0.515]	21600	21600	18636
DynEGA	Adjusted Rand index	0.902	0.0018	[0.899, 0.906]	21600	21600	18636
DynEGA	Variation of information	0.137	0.0025	[0.132, 0.143]	21600	21600	18636
DynEGA	Assignment accuracy given valid network output	0.935	0.0012	[0.933, 0.938]	21600	21600	18636
GraphicalVAR	Signed dimension error	1.048	0.0154	[1.017, 1.076]	21579	21579	16641
GraphicalVAR	Absolute dimension error	1.048	0.0154	[1.016, 1.080]	21579	21579	16641
GraphicalVAR	Adjusted Rand index	0.859	0.0020	[0.855, 0.863]	21579	21579	16641
GraphicalVAR	Variation of information	0.200	0.0028	[0.195, 0.206]	21579	21579	16641
GraphicalVAR	Assignment accuracy given valid network output	0.893	0.0015	[0.890, 0.896]	21579	21579	16641
Static EGA	Signed dimension error	0.576	0.0136	[0.551, 0.603]	21541	21541	19721
Static EGA	Absolute dimension error	0.576	0.0136	[0.550, 0.603]	21541	21541	19721
Static EGA	Adjusted Rand index	0.922	0.0018	[0.919, 0.926]	21541	21541	19721
Static EGA	Variation of information	0.108	0.0025	[0.103, 0.113]	21541	21541	19721
Static EGA	Assignment accuracy given valid network output	0.943	0.0013	[0.940, 0.945]	21541	21541	19721

Table C7: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 100$. Correct-dimensional rows include valid outputs that recovered the correct number of dimensions. All-attempt failure-adjusted rows assign correlation 0 and NRMSE 2 when no valid network output was returned. These are post-estimation summaries of recovered groups; higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid	Correct dimension
DynEGA	All-attempt failure-adjusted trajectory correlation	0.549	0.0014	[0.546, 0.552]	21600	21600	20282
DynEGA	All-attempt failure-adjusted trajectory NRMSE	1.435	0.0013	[1.432, 1.438]	21600	21600	20282
GraphicalVAR	All-attempt failure-adjusted trajectory correlation	0.668	0.0020	[0.664, 0.671]	21600	20329	16784
GraphicalVAR	All-attempt failure-adjusted trajectory NRMSE	0.753	0.0036	[0.746, 0.760]	21600	20329	16784
GIMME	All-attempt failure-adjusted trajectory correlation	0.132	0.0019	[0.129, 0.136]	21600	4846	3795
GIMME	All-attempt failure-adjusted trajectory NRMSE	1.749	0.0034	[1.742, 1.756]	21600	4846	3795
Static EGA	All-attempt failure-adjusted trajectory correlation	0.727	0.0017	[0.723, 0.730]	21600	21526	19093
Static EGA	All-attempt failure-adjusted trajectory NRMSE	0.641	0.0031	[0.635, 0.647]	21600	21526	19093
DynEGA	Trajectory correlation (correct-dimensional valid network outputs)	0.574	0.0013	[0.572, 0.577]	20282	21600	20282
DynEGA	Trajectory NRMSE (correct-dimensional valid network outputs)	1.418	0.0014	[1.416, 1.421]	20282	21600	20282
GraphicalVAR	Trajectory correlation (correct-dimensional valid network outputs)	0.791	0.0014	[0.789, 0.794]	16784	20329	16784
GraphicalVAR	Trajectory NRMSE (correct-dimensional valid network outputs)	0.568	0.0031	[0.562, 0.574]	16784	20329	16784
GIMME	Trajectory correlation (correct-dimensional valid network outputs)	0.606	0.0034	[0.600, 0.613]	3795	4846	3795
GIMME	Trajectory NRMSE (correct-dimensional valid network outputs)	0.898	0.0050	[0.888, 0.907]	3795	4846	3795
Static EGA	Trajectory correlation (correct-dimensional valid network outputs)	0.787	0.0014	[0.784, 0.789]	19093	21526	19093
Static EGA	Trajectory NRMSE (correct-dimensional valid network outputs)	0.563	0.0031	[0.557, 0.569]	19093	21526	19093

Table C8: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 25$. Correct-dimensional rows include valid outputs that recovered the correct number of dimensions. All-attempt failure-adjusted rows assign correlation 0 and NRMSE 2 when no valid network output was returned. These are post-estimation summaries of recovered groups; higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid	Correct dimension
DynEGA	All-attempt failure-adjusted trajectory correlation	0.611	0.0011	[0.608, 0.613]	21600	21600	18636
DynEGA	All-attempt failure-adjusted trajectory NRMSE	1.412	0.0012	[1.409, 1.414]	21600	21600	18636
GraphicalVAR	All-attempt failure-adjusted trajectory correlation	0.799	0.0012	[0.796, 0.801]	21600	21579	16641
GraphicalVAR	All-attempt failure-adjusted trajectory NRMSE	0.587	0.0023	[0.582, 0.591]	21600	21579	16641
GIMME	All-attempt failure-adjusted trajectory correlation	0.000	0.0000	[0.000, 0.000]	21600	0	0
GIMME	All-attempt failure-adjusted trajectory NRMSE	2.000	0.0000	[2.000, 2.000]	21600	0	0
Static EGA	All-attempt failure-adjusted trajectory correlation	0.801	0.0012	[0.799, 0.803]	21600	21541	19721
Static EGA	All-attempt failure-adjusted trajectory NRMSE	0.595	0.0024	[0.591, 0.600]	21600	21541	19721
DynEGA	Trajectory correlation (correct-dimensional valid network outputs)	0.654	0.0010	[0.652, 0.655]	18636	21600	18636
DynEGA	Trajectory NRMSE (correct-dimensional valid network outputs)	1.382	0.0013	[1.380, 1.385]	18636	21600	18636
GraphicalVAR	Trajectory correlation (correct-dimensional valid network outputs)	0.871	0.0008	[0.870, 0.873]	16641	21579	16641
GraphicalVAR	Trajectory NRMSE (correct-dimensional valid network outputs)	0.466	0.0022	[0.461, 0.470]	16641	21579	16641
Static EGA	Trajectory correlation (correct-dimensional valid network outputs)	0.836	0.0009	[0.834, 0.838]	19721	21541	19721
Static EGA	Trajectory NRMSE (correct-dimensional valid network outputs)	0.548	0.0023	[0.544, 0.553]	19721	21541	19721

C.3 Targeted Departures, Between-Clip Analyses, and Scenario Parameters

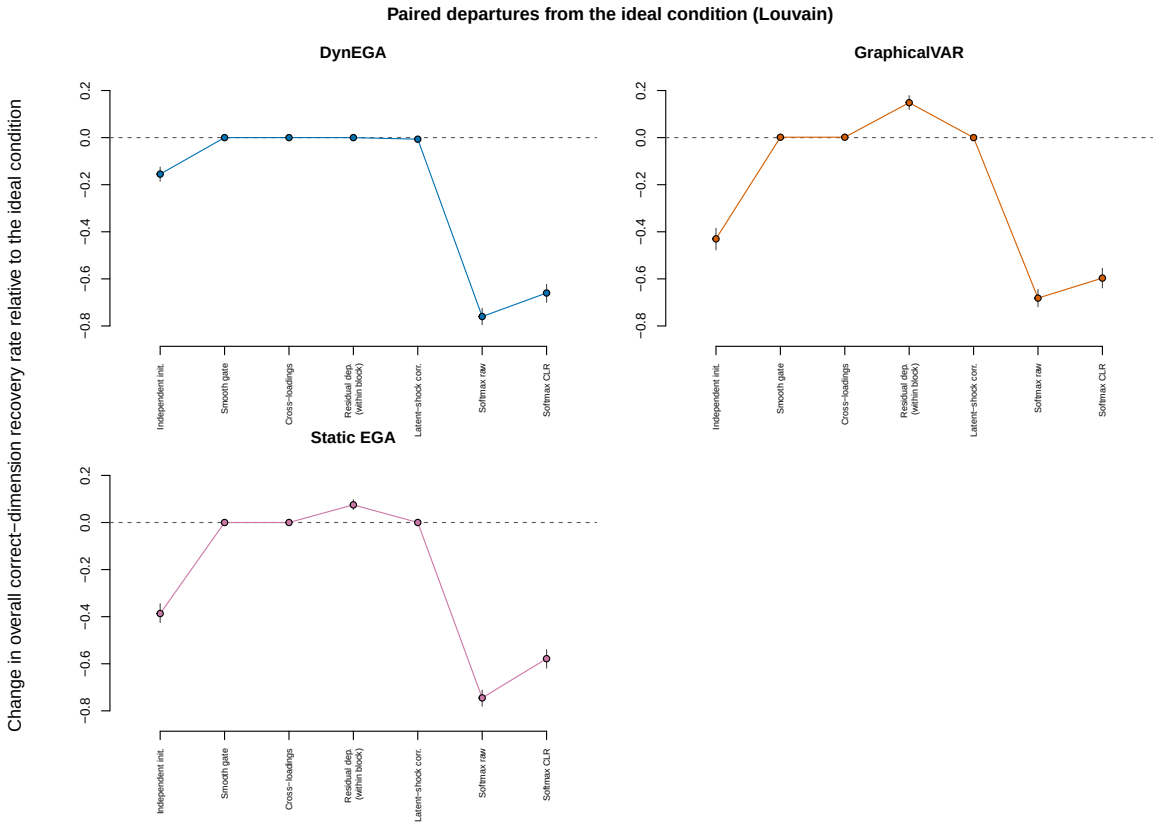


Figure C3: Paired change in overall correct-dimension recovery under each targeted departure. Negative values indicate lower recovery than in the fixed equal-measurement condition. Intervals resample whole datasets.

The scenario families use the numeric settings below. The displayed σ_q vectors are the per-step acceleration-input standard deviations used in the analyses. The primary simulation design with 100 observations uses $\Delta = 0.4$. Conditions with 300 observations occur only in the extended factorial simulation and use $\Delta = 0.25$, so these conditions differ in the number of observations, the interval between them, the total time spanned, and discrete acceleration-input specification.

Codes A–D use asymptotically stable feedback. Codes E–F use $\eta_f < 0$ and $\zeta_f > 0$, which produces oscillatory finite-horizon amplification. Their interpretation is restricted to the stated observation horizons. The simulated-versus-real comparison uses stable negative-ζ dynamics and is analyzed separately. All oscillatory factors satisfy the cadence restriction $0 < \omega_f < \pi/\Delta$ at the observation intervals used here.

Table C9: Generator-only amplitude diagnostics for the finite-horizon amplification families with $N = 50$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and 25 replications. The final-tenth energy fraction is the proportion of squared activity-scaled latent signal occurring in the last 10% of observations, averaged across clips within a panel and then summarized across replications.

Scenario	Dimensions	T	H	Median max $\lvert\tilde{x}\rvert$	95th percentile max $\lvert\tilde{x}\rvert$	Median observed IQR	Final-tenth energy fraction
Ranked finite-horizon amplification	2	100	39.60	55.9	61.5	3.47	0.316
Ranked finite-horizon amplification	2	300	74.75	329.4	376.9	7.46	0.338
Ranked finite-horizon amplification	3	100	39.60	62.6	73.0	1.90	0.368
Ranked finite-horizon amplification	3	300	74.75	430.3	478.6	4.58	0.428
Ranked finite-horizon amplification	4	100	39.60	85.6	102.7	1.57	0.404
Ranked finite-horizon amplification	4	300	74.75	994.9	1092.7	4.02	0.498
Switching finite-horizon amplification	2	100	39.60	69.5	81.9	4.71	0.319
Switching finite-horizon amplification	2	300	74.75	388.1	491.3	9.31	0.324
Switching finite-horizon amplification	3	100	39.60	77.7	87.5	2.33	0.367
Switching finite-horizon amplification	3	300	74.75	518.5	608.5	5.74	0.448
Switching finite-horizon amplification	4	100	39.60	105.2	122.2	1.92	0.402
Switching finite-horizon amplification	4	300	74.75	1201.6	1350.2	4.97	0.517

Table C10: Activity schedules and dynamic-feedback settings for each scenario family.

Code	Family	Dynamic family	Switching	Gain	Dwell	Velocity feedback
A	Low-amplitude single dominance	Stable template	No	0.35 / 0.05	1.00 / 1.00	Negative
B	High-amplitude single dominance	Stable template	No	1.00 / 0.08	1.00 / 1.00	Negative
C	High-amplitude switching	Stable template	Yes	1.00 / 0.08	0.20 / 0.45	Negative
D	High-amplitude rapid switching	Stable template	Yes	1.25 / 0.08	0.03 / 0.10	Negative
E	Ranked finite-horizon amplification	Amplification	No	1.00 / 0.08	1.00 / 1.00	Positive
F	Switching finite-horizon amplification	Amplification	Yes	1.10 / 0.10	0.12 / 0.30	Positive

Table C11: Dynamic parameters for each scenario family. Acceleration-input SD is the per-step standard deviation σ_q used in each condition.

Code	Dimensions	eta	zeta	Acceleration-input SD
A	2D	-0.30; -0.55	-1.70; -1.00	0.06; 0.15
A	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.06; 0.15; 0.23
A	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.06; 0.15; 0.23; 0.33
B	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
B	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
B	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
C	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
C	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
C	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
D	2D	-0.30; -0.55	-1.70; -1.00	0.15; 0.38
D	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.15; 0.38; 0.56
D	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.15; 0.38; 0.56; 0.81
E	2D	-0.05; -0.10	0.08; 0.10	0.25; 0.25

E	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.25; 0.25; 0.25
E	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.25; 0.25; 0.25; 0.25
F	2D	-0.05; -0.10	0.08; 0.10	0.35; 0.35
F	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.35; 0.35; 0.35
F	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.35; 0.35; 0.35; 0.35

Table C12: Deterministic stability diagnostics for the unique dynamic-parameter pairs used across scenario factors. Table C11 maps these pairs to scenarios and factors. The modal exponential-rate factors $g(H) = \exp\{H \max \mathrm{Re}(r)\}$ describe modal rates over the observed horizon rather than total finite-horizon state amplification. $H[100] = 39.6$ and $H[300] = 74.75$ under the stated record designs.

Scenario codes	eta	zeta	Discriminant	Max Re(root)	rho(Phi[.4])	rho(Phi[.25])	g(H[100])	g(H[300])	Regime
A, B, C, D	-1.10	-0.30	-4.3100	-0.150	0.9418	0.9632	0.0026	0.0000	Stable oscillatory
A, B, C, D	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
A, B, C, D	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
A, B, C, D	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
E, F	-0.20	0.14	-0.7804	0.070	1.0284	1.0177	15.9906	187.2604	Finite-horizon amplifying oscillatory
E, F	-0.15	0.12	-0.5856	0.060	1.0243	1.0151	10.7618	88.6770	Finite-horizon amplifying oscillatory
E, F	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
E, F	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory

C.4 Extended Factorial Simulation

The extended factorial simulation evaluated DynEGA on 518,400 datasets. It recovered the correct dimension in 93.7% of those datasets and assigned 95.2% of indicators correctly. Stronger loadings and smaller residual error improved recovery. The design, including the $T = 300$ change in observation interval, is given in Appendix B. Runtime accounting for this design is in Section C.5.

C.5 Runtime and Feasibility Accounting

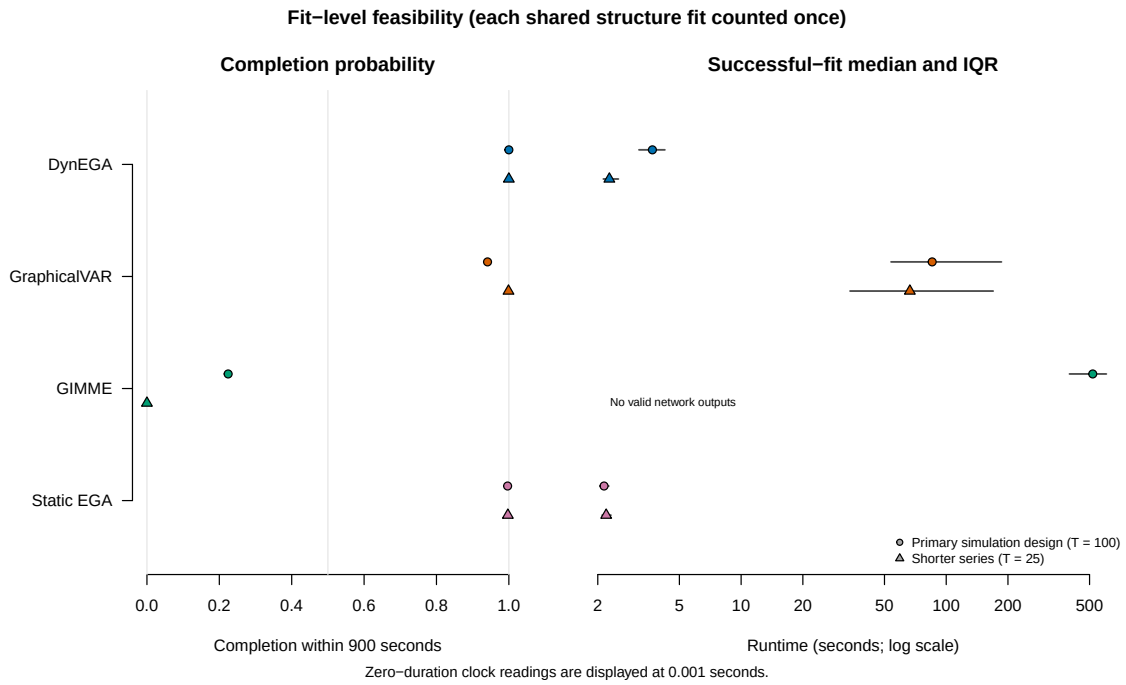


Figure C4: Runtime among valid network outputs in the primary $T = 100$ and shorter $T = 25$ designs. Points show medians and intervals show interquartile ranges on a log10 scale. Counts use unique fitted structures.

Table C13: Runtime and feasibility per method fit in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Completion intervals resample whole simulated datasets. Runtime summaries condition on valid network outputs and include structure estimation, network extraction, and community detection. Each fitted structure is reused for Louvain, Leiden, and Walktrap.

Analysis	Time-series length	Method	Attempts	Valid outputs	Timeouts	Errors	Completion	Runtime median [IQR], s
Matched-method simulation design	$T = 100$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	3.69 [3.18, 4.25]
Matched-method simulation design	$T = 100$	GIMME	21600	4846	14477	2277	0.224 [0.219, 0.230]	518.16 [398.98, 603.71]
Matched-method simulation design	$T = 100$	GraphicalVAR	21600	20329	1271	0	0.941 [0.938, 0.944]	85.41 [53.82, 185.91]
Matched-method simulation design	$T = 100$	Static EGA	21600	21526	0	74	0.997 [0.996, 0.997]	2.15 [2.05, 2.26]
Matched-method simulation design	$T = 25$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	2.28 [2.13, 2.52]
Matched-method simulation design	$T = 25$	GIMME	21600	0	0	21600	0.000 [0.000, 0.000]	Not available
Matched-method simulation design	$T = 25$	GraphicalVAR	21600	21579	21	0	0.999 [0.999, 0.999]	66.49 [33.96, 169.39]
Matched-method simulation design	$T = 25$	Static EGA	21600	21541	0	59	0.997 [0.997, 0.998]	2.20 [2.10, 2.32]

C.6 Community-Detection Sensitivity

Louvain, Leiden, and Walktrap were applied to each retained fitted graph. Their results were closely aligned in the matched-method simulation designs (Figure C5) and led to the same interpretation of the targeted sensitivity conditions (Figure C6). These comparisons reuse fitted structures and therefore assess the community-detection step rather than estimator variability.

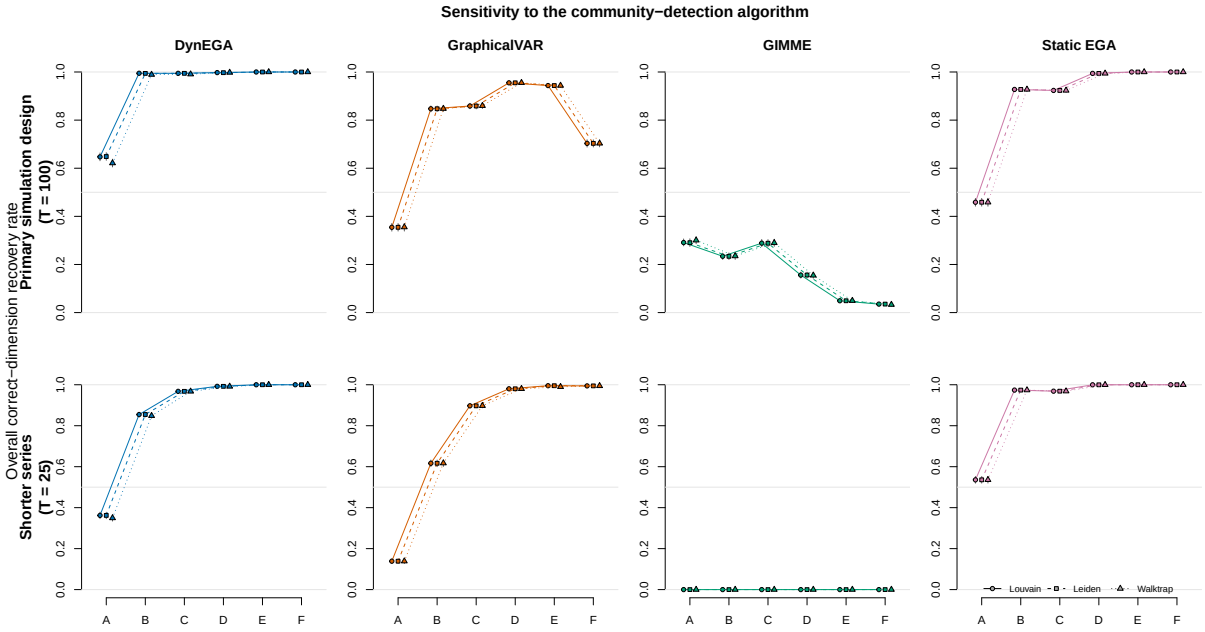


Figure C5: Overall correct-dimension recovery rates for Louvain, Leiden, and Walktrap in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Each fitted network is reused across all three algorithms, so differences isolate the community-detection step.

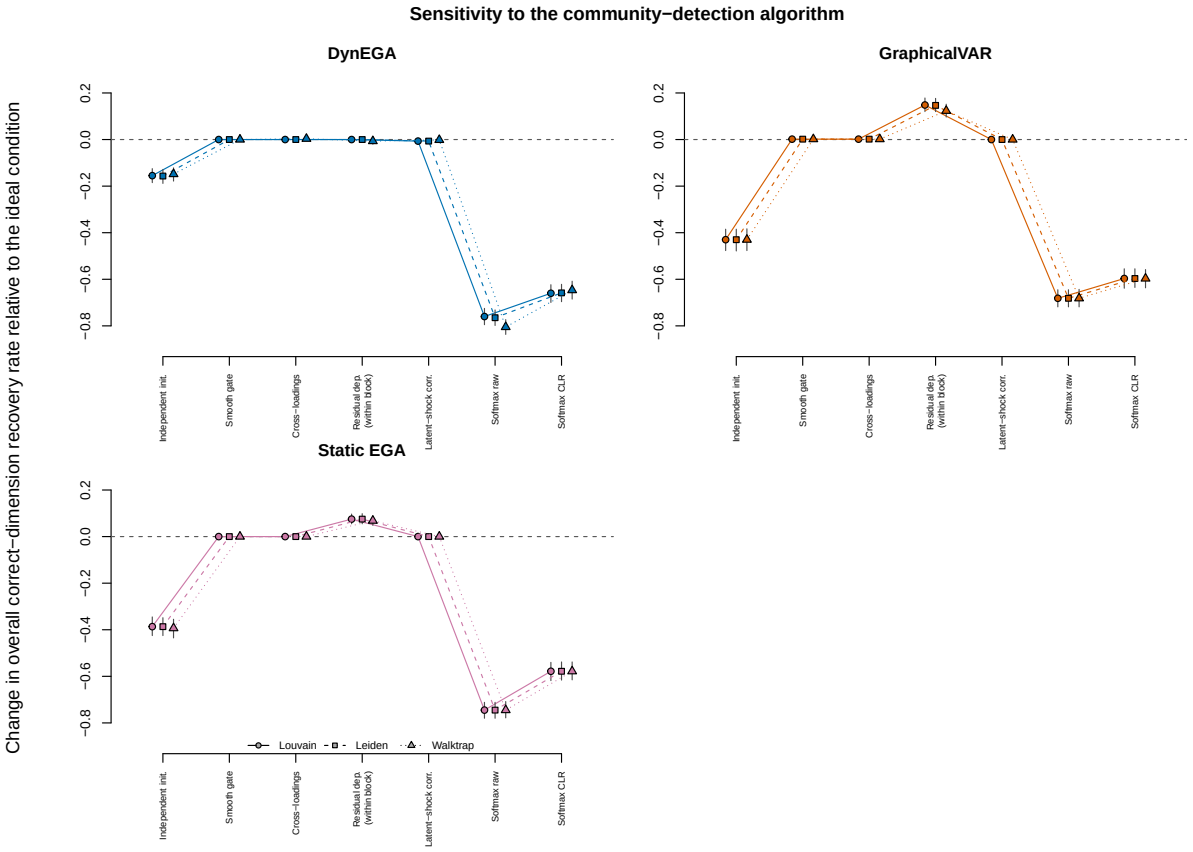


Figure C6: Sensitivity of the paired departure effects to Louvain, Leiden, and Walktrap. Values are changes in overall correct-dimension recovery rate relative to the ideal condition.